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COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

CAPSTONE PROJECT PROPOSAL

**"WEB-BASED MR. SOFTY INTEGRATED BUSINESS INTELLIGENCE
PLATFORM"**

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ABSTRACT

This study focused on the design and development of a **Web-Based Mr. Softy Integrated Business Intelligence Platform** intended to improve the efficiency and accuracy of the business's daily operations. Mr. Softy currently relies on manual processes for sales recording, inventory monitoring, and employee attendance tracking, which often result in delays, inaccuracies, and difficulties in managing business records. To address these challenges, the researchers proposed a centralized web-based system that integrates sales management, inventory management, attendance monitoring, and report generation into a single platform.

The study employed a developmental research design and utilized the Agile Software Development Methodology to guide the system development process. Data were gathered through interviews and observations conducted with the business owner and staff to identify operational problems and system requirements. The system was developed using HTML, CSS, JavaScript, PHP, and MySQL, following a client-server architecture to support efficient data processing and management.

The proposed platform enables automated recording of sales transactions, real-time inventory monitoring, employee attendance tracking, and generation of business reports. The system was designed to reduce manual workloads, minimize human errors, improve data organization, and provide accurate information that supports business decision-making. Furthermore, the platform offers secure access for authorized users and a user-friendly interface that promotes ease of use and operational efficiency.

The study concludes that the implementation of a web-based integrated business management platform can significantly enhance the operational performance of Mr. Softy by providing a more efficient, accurate, and organized approach to managing sales, inventory, and attendance records. Future enhancements may include mobile application support, advanced analytics, online transaction capabilities, and cloud-based deployment to further improve system functionality and business performance.

Keywords: Business Intelligence Platform, Web-Based System, Sales Management, Inventory Management, Attendance Monitoring, Agile Methodology, Business Automation.

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CHAPTER I

INTRODUCTION

In today's digital era, technology plays an important role in improving the efficiency and productivity of businesses. Many companies and establishments are now adopting computerized systems to make their operations faster, more accurate, and more organized. Through the use of technology, businesses can easily manage transactions, monitor inventory, track employee attendance, and generate reports that help in decision-making. Compared to manual methods, computerized systems provide better data management, reduce human errors, and improve the overall workflow of the business.

For small and medium-sized businesses, the use of information technology is becoming more important as business operations continue to grow. Manual processes may still be effective for small-scale operations, but they often become inefficient and difficult to manage when the volume of transactions increases. Traditional methods such as writing sales records in notebooks, manually checking inventory, and using logbooks for attendance tracking can consume a lot of time and may lead to inaccurate records, lost data, and delayed transactions. These problems can affect the performance and productivity of the business.

One of the common problems experienced by businesses that still rely on manual processes is inaccurate record management. Manual sales recording may result in incorrect computations and incomplete reports, while poor inventory monitoring can lead to overstocking or shortages of products. In addition, manual

attendance tracking may cause errors in employee work records and difficulty in monitoring employee performance. Because of these challenges, businesses need a more

efficient and reliable system that can automate their daily operations and provide accurate information in real time.

Mr. Softy, a local business, currently uses manual methods in managing sales transactions, employee attendance, and inventory monitoring. The business records sales manually, monitors inventory without an automated system, and tracks employee attendance using traditional methods. As a result, the business experiences difficulties in organizing records, retrieving information quickly, and maintaining accurate data. These challenges may slow down business operations and increase the possibility of errors, especially as the business continues to expand.

To address these problems, this study proposes the development of a **Web-Based Mr. Softy Integrated Business Intelligence Platform**. The proposed system aims to automate important business processes such as sales recording, attendance tracking, inventory management, and report generation. The system will provide a centralized platform where business data can be stored, processed, and monitored efficiently. Through this platform, the business owner can access real-time information, minimize manual work, reduce errors, and improve decision-making.

The development of the proposed system is expected to help Mr. Softy improve the efficiency and accuracy of its daily operations. Furthermore, the study aims to demonstrate how web-based technologies and business intelligence solutions can help small businesses adapt to modern digital practices and achieve better operational management.

Statement of the Problem

Mr. Softy currently relies on manual methods for managing sales transactions, employee attendance, and inventory records. Although these methods are manageable for small-scale operations, they often result in inefficiencies such as time-consuming processes, inaccurate record-keeping, data redundancy, and difficulty in retrieving information when needed. These challenges can affect the overall productivity and decision-making capabilities of the business.

In addition, the absence of an integrated system causes business information to be stored separately, making it difficult for the owner to effectively monitor daily operations, track inventory levels, manage employee attendance, and generate accurate reports. As the business continues to operate and expand, the need for a computerized and centralized management system becomes increasingly important.

This study aims to develop an Integrated Sales, Inventory, and Attendance Management System for Mr. Softy. Specifically, it seeks to answer the following questions:

1. How can sales transactions be automated to improve accuracy, reduce human errors, and streamline the recording of daily sales?
2. How can employee attendance be monitored and recorded efficiently to ensure accurate attendance tracking and reporting?
3. How can inventory be managed effectively to maintain accurate stock records and prevent stock shortages or overstocking?
4. How can the integration of sales, inventory, and attendance management into a single system improve the efficiency of business operations?

5. How can the proposed system provide timely and accurate reports that support the business owner in making informed decisions?

Objectives of the Study

General Objective

The primary objective of this study is to design, develop, and implement a web-based Sales, Attendance, and Inventory Management System for Mr. Softy to automate and improve its daily business operations. The proposed system aims to replace the existing manual processes with a computerized platform that can efficiently manage sales transactions, employee attendance records, and inventory information within a single integrated system.

Furthermore, the study seeks to provide a more accurate, reliable, and organized method of recording and storing business data, reducing the risk of errors, data loss, and inconsistencies commonly associated with manual record-keeping. Through automation, the system is expected to streamline operational tasks, minimize the time required for data processing, and improve the accessibility of important business information.

The proposed system will also enable the business owner and administrators to monitor sales performance, employee attendance, and inventory levels in real time, allowing them to make informed decisions based on accurate and up-to-date data. By centralizing these essential business functions, the system aims to enhance productivity, improve workflow efficiency, strengthen data management practices, and support the overall growth and operational effectiveness of Mr. Softy.

Specific Objectives

Specifically, this study aims to achieve the following objectives:

1. To develop a sales management module that can efficiently record, process, and manage daily sales transactions, automatically calculate totals, and securely store transaction records to enhance the accuracy and reliability of sales operations.
2. To create an attendance monitoring module that can accurately record employee time-in and time-out, enabling real-time attendance tracking and reducing errors in monitoring employee work hours.
3. To design and implement an inventory management module that can monitor product availability, automatically update stock levels based on transactions, and help prevent inventory-related issues such as stock shortages, overstocking, and inaccurate inventory records.
4. To minimize human errors, delays, and inconsistencies associated with manual recording and data management processes through the implementation of a computerized and automated system.
5. To provide organized, secure, and easily accessible records of sales, inventory, and employee attendance that can support effective monitoring, reporting, and decision-making by the business owner and administrators.
6. To integrate sales management, inventory control, and attendance monitoring into a centralized web-based system that streamlines business operations and improves overall workflow efficiency.
7. To develop a user-friendly and intuitive interface that enables staff and administrators to efficiently perform their tasks with minimal training and difficulty.

8. To generate accurate and timely reports on sales performance, inventory status, and employee attendance that can assist management in evaluating business operations and planning future actions.
9. To evaluate the developed system in terms of functionality, usability, reliability, efficiency, and security to ensure that it meets the operational requirements of the business.

Scope and Delimitation

Scope of the Study

This study focuses on the design and development of a web-based Sales, Attendance, and Inventory Management System for Mr. Softy. The proposed system aims to provide a computerized solution that will help improve the efficiency, accuracy, and organization of the business's daily operations. The system is intended to replace the traditional manual process of recording sales, monitoring inventory, and tracking employee attendance, which are often time-consuming and prone to human error. Through the implementation of the system, the business will be able to manage records more effectively and improve overall operational performance.

The proposed system will be accessible to authorized users, specifically the administrator and staff members of Mr. Softy. Each user will have designated access to the system depending on their assigned role and responsibilities. The administrator will be responsible for managing products, monitoring inventory, viewing reports, and overseeing system operations, while staff members will be able to record customer orders, process transactions, check stock availability, and log attendance. The use of user authentication and login credentials will help ensure the security and confidentiality of business data stored in the system.

The system will include several major features necessary for the daily operations of the business. For sales management, the system will allow users to record customer orders, calculate totals automatically, process billing, and save transaction records for future reference. For inventory management, the system will monitor product stocks, update

inventory levels whenever sales are made, and help track the availability of products to avoid shortages or overstocking. In terms of attendance management, the system will record employee time-in and time-out logs to provide accurate attendance monitoring and maintain organized employee records.

Additionally, the system will include a basic reporting feature that can generate summaries of sales, attendance, and inventory records. These reports will help the business owner and administrators monitor business performance and make informed decisions based on accurate data. The proposed system will be web-based, allowing it to run through a web browser using computers or other compatible devices connected to the internet or local network. This setup will make the system more accessible, convenient, and easier to maintain compared to traditional paper-based methods.

Overall, the study focuses on developing a functional and user-friendly management system that will improve the workflow of Mr. Softy by automating important business processes. The researchers aim to create a system that can reduce manual workloads, minimize recording errors, improve data organization, and contribute to better business operations and customer service.

Delimitation of the Study

This study is limited only to the development of a web-based Sales, Attendance, and Inventory Management System intended for the internal operations of Mr. Softy. The proposed system will focus mainly on managing sales transactions, employee attendance, and inventory monitoring within the business establishment. The study does not cover operations beyond the identified functions and objectives of the system.

The system will not include features related to online shopping, customer accounts, online reservations, online payment processing, delivery services, or integration with third-party applications. Customers will not have direct access to the system since it is intended solely for use by administrators and staff members of the business. The study is also limited to basic report generation and does not include advanced financial management functions such as accounting systems, payroll computation, tax management, or detailed business analytics.

Furthermore, the proposed system will not include mobile application support, artificial intelligence capabilities, automated forecasting, or cloud-based enterprise deployment. The researchers will only develop the system using the available tools, programming languages, resources, and timeframe allotted for the study. Because of these limitations, some advanced functionalities that may be present in commercial business management systems will not be implemented in the proposed system. The implementation and testing of the system will only be conducted within the business environment of Mr. Softy.

The performance and effectiveness of the system may vary when applied to larger businesses or different operational setups. Future enhancements, additional modules, and system upgrades beyond the scope identified in this study are not included and may be recommended for further development by future researchers.

Significance of the Study

The results of this study are expected to provide significant benefits to the business, its employees, customers, and future researchers. The development of a web-based Sales, Attendance, and Inventory Management System for Mr. Softy aims to improve the efficiency, accuracy, and organization of business operations. Specifically, this study will benefit the following:

Mr. Softy Business

The proposed system will help Mr. Softy improve its daily operations by automating sales recording, employee attendance tracking, and inventory management. Through the use of a computerized system, the business can reduce manual workloads, minimize recording errors, and organize business data more effectively. The system is also expected to improve operational efficiency, save time, and provide a more reliable method of handling important business information.

Business Owner and Management

The business owner and management will benefit from the system through easier access to organized and accurate records related to sales, inventory, and employee attendance. The system will provide real-time monitoring and basic reports that can support decision-making and business planning. With accurate and updated information, management can monitor business performance more effectively, identify operational issues quickly, and make informed decisions that may contribute to the growth and improvement of the business.

Employees

The proposed system will help employees by providing a more convenient and accurate method of recording attendance and managing work-related tasks. The attendance monitoring feature will reduce errors in recording work hours and help ensure that employee attendance records are properly documented. In addition, the system can simplify certain operational tasks, making daily work more organized and efficient for staff members.

Customers

Although the system is mainly intended for internal business operations, customers may also benefit indirectly from the improved efficiency of the business. Faster transaction processing, more accurate order handling, and better inventory monitoring can contribute to improved customer service and smoother business transactions. The system may also help reduce delays and mistakes during customer transactions, leading to a better overall customer experience.

Future Researchers

This study may serve as a useful reference and guide for future researchers who plan to conduct studies related to sales systems, attendance monitoring systems, inventory management systems, or other business-related information systems. The findings, design, and development process of this study may provide valuable insights and additional knowledge that can help future researchers improve or develop similar systems in different business environments.

Definition of Terms

The following terms are operationally and conceptually defined to provide better understanding of the study and the proposed system:

1. **Authorized User-** Refers to an individual who is granted permission to access and use the system based on assigned roles and responsibilities. In this study, authorized users include the administrator and staff members of Mr. Softy.
2. **Business Intelligence Platform-** Refers to a system or technology used to collect, analyze, and present business data in order to support effective decision-making and improve business operations.
3. **Computerized Management System-** Refers to a digital or computer-based system designed to manage, organize, process, and store business information more efficiently compared to manual methods.
4. **Employee Attendance Tracking-** Refers to the process of recording, monitoring, and managing employee attendance, including time-in and time-out records, to ensure accurate documentation of work hours.
5. **Inventory Management-** Refers to the process of monitoring, organizing, and controlling product stocks to maintain accurate inventory records and avoid shortages or overstocking of products.
6. **Manual System-** Refers to a traditional method of handling business operations and records using paper-based processes or manual encoding without the assistance of automated computerized systems.
7. **Product Inventory-** Refers to the collection of products, goods, or items available for sale or use within the business establishment.

8. **Sales Recording-** Refers to the process of documenting and storing sales transactions, including customer orders, purchased products, quantities, and total amounts.

9. **Stock Monitoring-** Refers to the activity of checking, updating, and tracking the quantity and availability of products in the inventory to ensure proper stock management.

10. **System Administrator-** Refers to the authorized person responsible for managing the system, monitoring operations, maintaining records, and controlling user access within the proposed system.

11. **Web-Based System-** Refers to a computerized system that can be accessed through a web browser using an internet connection or local network, allowing users to perform operations online.

12. **Real-Time Monitoring-** Refers to the ability of the system to display updated information immediately as transactions, attendance logs, or inventory changes occur within the system.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies that support the development of the proposed web-based Sales, Attendance, and Inventory Management System for Mr. Softy. The researchers gathered information from books, journals, articles, online sources, and previous studies related to computerized systems, inventory management, attendance monitoring, and sales management. These related studies and literature provide ideas, concepts, and information that are helpful in understanding how technology can improve business operations and management.

Nowadays, many businesses use computerized systems to make their operations faster, easier, and more accurate. Traditional manual processes in recording sales, monitoring inventory, and tracking employee attendance are often time-consuming and prone to human errors. Because of this, many small and medium businesses are now adopting digital systems to improve efficiency, organize records properly, and reduce mistakes in handling business data.

Several studies show that web-based and computerized management systems help businesses improve transaction

processing, inventory monitoring, report generation, and employee management. Sales and inventory systems allow businesses to monitor products and transactions in real time, while attendance systems help ensure accurate recording of employee work hours.

These systems also help business owners access organized information that can support better decision-making and business monitoring.

The related literature and studies included in this chapter serve as the foundation of the proposed study. They help the researchers identify the strengths and weaknesses of existing systems and provide guidance in designing and developing the proposed system for Mr. Softy. Furthermore, the gathered information helps the researchers understand the importance of integrating sales, attendance, and inventory management into one centralized web-based system that can improve the overall efficiency and productivity of the business.

Foreign Studies

POINT OF SALES

The study conducted by the researchers (2025) focused on the development of an ASN disciplinary monitoring dashboard integrated within the SIPP application in Bondowoso Regency, Indonesia. The purpose of the study was to provide a more efficient, accurate, and integrated attendance monitoring system that supports employee discipline management and organizational supervision. The system was developed using the Agile Software Development Life Cycle (SDLC) methodology through continuous collaboration between developers, government agencies, and system operators.

The findings revealed that the dashboard significantly improved attendance monitoring through features such as attendance recaps, flexible data filtering, informative visualizations, and role-based access controls. The system provided different access levels for administrators, operators, and employees, enabling effective monitoring and management of attendance records. The researchers found that the dashboard improved monitoring efficiency and supported data-driven decision-making processes.

Furthermore, usability evaluations involving more than 500 users indicated a high level of satisfaction regarding the system's ease of use and effectiveness in supporting employee performance monitoring. The system centralized attendance information and generated comprehensive reports that enhanced transparency, accountability, and administrative efficiency. The researchers also observed that the dashboard facilitated systematic supervision of employee discipline through organized and accessible attendance data.

The researchers concluded that the attendance monitoring dashboard enhanced operational efficiency, attendance supervision, and employee discipline management. The study emphasized that integrated attendance monitoring systems contribute to better governance, transparency, and accountability while supporting organizational performance improvement.

This study is relevant to the present research because it highlights the importance of attendance monitoring dashboards, attendance reporting, role-based access management, and data-driven decision-making. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how attendance monitoring technologies can improve operational efficiency, employee accountability, and management effectiveness.

Asrani et al. (2024) discussed the importance of Point of Sale (POS) systems as integrated solutions that combine hardware and software to facilitate business transactions, inventory management, and sales monitoring. The study explained that modern POS systems enable businesses to record sales transactions, track inventory levels, process different payment methods, and generate reports that support decision-making. These features help improve operational efficiency, accuracy, and overall business performance.

The authors further emphasized that POS systems play a significant role in inventory management through real-time stock monitoring, inventory updates, and automated inventory control processes. Such functionalities help businesses maintain accurate inventory records, prevent stock shortages, and improve inventory planning. The study also highlighted the integration capabilities of POS systems, allowing businesses to combine transaction processing with data analysis tools for better business insights.

Moreover, the research examined the feasibility of implementing web-based POS systems for small and medium enterprises (SMEs). The findings revealed that web-based systems provide advantages such as real-time access to information, improved sales tracking, efficient inventory management, and enhanced business monitoring. The study concluded that POS systems contribute to increased efficiency, improved customer service, better inventory control, enhanced security, and more accurate reporting, making them valuable tools for modern business operations.

This study is relevant to the present research because it supports the development of a web-based Sales, Attendance, and Inventory Management System for Mr. Softy. The features and benefits of POS systems discussed in the study demonstrate how computerized systems can improve business processes, data management, and operational efficiency.

According to Rahangdale, Raut, Ingle, and Vaidya (2024), the implementation of a Point of Sale (POS) Inventory Management System significantly improves the efficiency and accuracy of business operations. The study was conducted to develop a computerized system capable of managing sales transactions, inventory records, report generation, and user monitoring. The researchers aimed to address the limitations of traditional manual processes, which are often time-consuming, prone to errors, and difficult to manage as business operations expand.

The system was developed using the Agile Development Cycle, with C# and the .NET Framework as the primary programming tools and MySQL as the database management system. To assess the quality of the developed system, the researchers utilized the ISO 25010 software quality standards. The evaluation results showed that the system effectively streamlined inventory management and sales processing while providing accurate and reliable reports.

The findings revealed that the system improved inventory tracking through real-time updates, reduced data entry errors, enhanced operational efficiency, and provided valuable information to support business decision-making. Furthermore, the system enabled businesses to monitor inventory levels more effectively, minimize stock shortages and overstocking, and improve customer service through faster transaction processing.

The study concluded that a POS Inventory Management System plays a vital role in modern business operations by automating routine processes and improving overall productivity. The researchers emphasized that computerized systems provide businesses with accurate data, real-time monitoring capabilities, and efficient reporting tools that contribute to better operational performance and organizational growth.

This study is relevant to the present research because it demonstrates the effectiveness of integrating sales and inventory management functions into a computerized system. Similar to the proposed Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights how automation can improve operational efficiency, ensure accurate record management, and support business decision-making through organized and reliable information processing.

Ramadhani, Nindyasari, and Murti (2025) conducted a study entitled “*Design and Development of a Web-Based Point of Sale System for Small-Scale Retail Management*” in Indonesia. The study aimed to address inventory inaccuracies, sales record discrepancies, and operational inefficiencies caused by manual record-keeping practices in small-scale retail businesses. The researchers developed a web-based Point of Sale (POS) system integrated with inventory management, sales recording, financial monitoring, and reporting functions to improve business operations and customer service.

The study utilized a mixed-methods research approach, combining qualitative and quantitative techniques to identify user requirements and evaluate the effectiveness of the developed system. The system was developed using PHP, MySQL, CodeIgniter, and Bootstrap and followed a prototyping development model with a user-centered design approach. System testing was conducted through functional and black-box testing to ensure that all features performed according to the specified requirements.

The findings revealed that the implementation of the web-based POS system significantly improved the accuracy of inventory records, reduced operational errors, enhanced sales monitoring, and streamlined business processes. The system also provided real-time inventory updates, improved data accessibility, and supported faster and more accurate decision-making. Furthermore, the integration of inventory and sales management functions enabled the business to optimize resource utilization and improve overall operational efficiency.

The researchers concluded that web-based POS systems provide an effective solution for small and medium-sized enterprises (SMEs) by improving inventory control, sales management, and business productivity. The study emphasized that adopting digital technologies can help businesses reduce manual errors, improve service quality, and remain competitive in an increasingly digital business environment.

This study is relevant to the present research because it demonstrates the effectiveness of a web-based system in managing sales transactions and inventory records. Similar to the proposed Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights the importance of automation in improving operational efficiency, ensuring data accuracy, and providing reliable information for business decision-making. The findings support the development of an integrated management system that can streamline daily operations and enhance overall business performance.

According to Prudente, Crodua, Adrian, Ramirez, and Romero (2026), the integration of Point-of-Sale (POS) and Financial Reporting Systems significantly improves retail performance, operational efficiency, and customer satisfaction. The study aimed to determine how the integration of sales processing and financial reporting contributes to accurate record management, faster transaction processing, improved inventory control, and better business decision-making in retail organizations.

The researchers employed a quantitative-descriptive research design and evaluated the developed system using the System Usability Scale (SUS). The study involved retail managers and finance personnel who assessed the system in terms of functionality, accuracy, reliability, and user acceptability. The results revealed an overall SUS score of 77.17, interpreted as “Good to Excellent Usability,” indicating that the system effectively supported retail operations and financial management.

The findings showed that the integrated POS and Financial Reporting System enhanced transaction speed, improved inventory accuracy, reduced human errors, streamlined report generation, and provided real-time business information. The system also improved decision-making by offering accurate financial reports and performance analytics that enabled managers to monitor sales, inventory levels, and operational performance efficiently. Furthermore, the integration of sales and financial data promoted transparency, consistency, and customer satisfaction through improved service quality and faster transaction processing.

The study concluded that integrated POS and Financial Reporting Systems play a vital role in modern retail management by automating business processes, improving data accuracy, and supporting informed decision-making. The researchers emphasized that digital business solutions contribute to operational efficiency, financial transparency, and overall organizational productivity.

This study is relevant to the present research because it supports the development of a Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy. Similar to the proposed system, the study highlights the importance of integrating sales monitoring, inventory management, and reporting functions into a computerized platform. The findings demonstrate how automation can improve operational efficiency, ensure accurate record management, and provide reliable information for business decision-making.

According to Umar, Zwal, Hussaini, and Oluyemisi (2021), Point-of-Sale (POS) systems play a significant role in improving business performance, operational efficiency, and sales management within small-scale businesses. The study entitled “Effect of Point of Sale (POS) on the Growth of Business in Nigeria” was conducted to examine the impact of POS systems on selected supermarkets in Bauchi Metropolis, Nigeria, and to determine the level of POS adoption among small businesses.

The researchers utilized a descriptive research design and gathered data through questionnaires administered to customers and business establishments. The collected data were analyzed using frequency counts and percentage distributions to determine customer perceptions, usage patterns, and the effectiveness of POS systems in business operations.

The findings revealed that POS systems provide businesses with efficient transaction processing, real-time inventory monitoring, and improved sales management. The study emphasized that POS technology enables retailers to access accurate management information, maintain better inventory control, reduce administrative workload, and support effective business decision-making. Furthermore, the results showed that 80% of the respondents agreed that POS systems contribute to increased sales, productivity, and business growth. The study also found that the adoption and usage of POS systems among retail businesses continue to increase due to their convenience and operational benefits.

The researchers concluded that POS systems are valuable technological tools that enhance retail business operations by improving transaction efficiency, supporting inventory management, and increasing organizational productivity. Although challenges such as network failures, power interruptions, and security concerns were identified, the benefits of POS systems were found to outweigh these limitations.

This study is relevant to the present research because it supports the development of a Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy. Similar to the proposed system, the study highlights the importance of automating sales transactions, monitoring inventory accurately, and improving operational efficiency through the use of information technology. The findings demonstrate how POS-based systems can contribute to better business performance, accurate record management, and improved decision-making.

ATTENDANCE MONITORING

The study conducted by Mohamad Noor, Ibrahim, and Jamaluddin (2023) focused on the development of an Employee Attendance Management System using Flutter technology. The purpose of the study was to provide a cost-effective and efficient attendance monitoring solution for organizations that still rely on manual attendance tracking methods. The researchers developed a mobile application that enables employees to record their attendance digitally while allowing administrators to monitor attendance records through a centralized management dashboard.

The findings revealed that the system improved attendance monitoring efficiency and reduced the administrative burden associated with manual attendance recording. The Flutter-based application provided cross-platform compatibility for Android and iOS devices, making the system accessible and convenient for employees and administrators. The researchers found that the system facilitated real-time attendance monitoring, report generation, and employee record management.

Furthermore, the system enhanced data accuracy and accessibility by maintaining attendance records within a centralized database. The researchers observed that the digital attendance process improved operational efficiency and minimized record-keeping errors commonly associated with traditional attendance methods. User evaluations also indicated a high level of acceptance and satisfaction with the system.

The researchers concluded that mobile-based attendance management systems provide an effective, reliable, and affordable solution for employee attendance monitoring. The study emphasized the importance of digital attendance technologies in improving workforce management and organizational productivity.

This study is relevant to the present research because it highlights employee attendance automation, centralized record management, and real-time reporting. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital attendance systems can improve efficiency, accuracy, and administrative effectiveness.

The study conducted by Nazara and Nasien (2024) focused on the development of an Employee Attendance System using the Rapid Application Development (RAD) method integrated with Location-Based Service (LBS) technology. The purpose of the study was to address the limitations of conventional attendance systems and provide a reliable solution for organizations implementing remote, field-based, or flexible work arrangements.

The findings revealed that the system successfully validated employee attendance using GPS-based location verification, ensuring that attendance could only be recorded within authorized locations. The web-based application enabled employees to submit attendance records digitally while administrators monitored attendance information through an integrated management platform. The researchers found that the system effectively improved attendance accuracy and accountability.

Furthermore, the implementation of the RAD methodology accelerated system development and enabled continuous improvements based on user feedback. Performance testing demonstrated that the application efficiently handled multiple simultaneous users while maintaining system responsiveness and reliability. The researchers observed that location verification significantly reduced attendance fraud and strengthened attendance monitoring procedures.

The researchers concluded that integrating location-based services into attendance management systems improves attendance validation, operational efficiency, and workforce accountability. The study recommended the adoption of location-aware attendance systems to support modern organizational requirements.

This study is relevant to the present research because it emphasizes web-based attendance management, GPS verification, and automated record keeping. Similar to the proposed

Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how location-based attendance technologies can improve monitoring accuracy, employee accountability, and administrative efficiency.

The study conducted by Srivathsan and Rajesh (2024) focused on the development of Work Track, an automated employee activity tracking and monitoring system designed to improve attendance recording and workforce management. The purpose of the study was to eliminate the inaccuracies and vulnerabilities associated with traditional manual attendance systems while providing organizations with a secure and efficient employee monitoring solution.

The findings revealed that the system significantly improved attendance accuracy through the implementation of digital clock-in and clock-out functionality integrated with GPS and geolocation verification. Employees were able to record attendance using a mobile application, while managers accessed real-time attendance information and performance reports through a centralized dashboard. The researchers found that the system reduced attendance discrepancies and enhanced workforce transparency.

Furthermore, the application incorporated additional features such as scheduling, break management, and activity monitoring, enabling organizations to manage employee operations more effectively. The researchers observed that the system improved data accessibility, streamlined attendance processes, and enhanced managerial decision-making through real-time reporting capabilities.

The researchers concluded that automated employee tracking and attendance systems provide a secure, efficient, and reliable solution for workforce management. The study emphasized the importance of integrating location verification and digital monitoring technologies to improve organizational productivity and accountability.

This study is relevant to the present research because it highlights automated attendance monitoring, employee activity tracking, GPS verification, and real-time reporting. Similar to

the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital workforce management systems can improve attendance accuracy, operational efficiency, and administrative control.

The study conducted by Munshi, Aljojo, Zainol, Al-Saadi, Babteen, Al-Hilal, Babader, Al-Zahrani, and Al-Abdulrahman (2019) focused on the development of an Employee Attendance Monitoring System by applying the concept of Enterprise Resource Planning (ERP). The purpose of the study was to improve employee attendance monitoring by integrating attendance records, human resource management, and administrative processes into a centralized web-based system. The researchers developed the system to automate attendance tracking and reduce the time and effort required in processing attendance records manually.

The findings revealed that the system significantly improved attendance monitoring efficiency by automatically calculating attendance delays and enforcing organizational attendance policies. The integration of the Human Resources Department, Finance Department, and administrative offices enabled seamless sharing of attendance information and facilitated faster decision-making processes. The researchers found that the system reduced errors associated with manual attendance recording and improved the accuracy of attendance reports.

Furthermore, the system generated various attendance-related reports, including attendance summaries, delay reports, absence records, and payroll-related deductions. The researchers observed that the centralized ERP-based approach improved transparency, accountability, and accessibility of attendance information. The system also reduced administrative workload and enabled managers to monitor employee attendance more effectively through automated reporting features.

The researchers concluded that the Employee Attendance Monitoring System enhanced operational efficiency, accuracy, and fairness in attendance management. The study

emphasized that integrating attendance monitoring with administrative and payroll functions provides organizations with a more effective and reliable method of managing employee attendance records.

This study is relevant to the present research because it highlights automated attendance monitoring, web-based record management, attendance reporting, and integration with organizational processes. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital attendance systems can improve efficiency, accountability, accuracy, and overall workforce management.

INVENTORY MANAGEMENT

The study conducted by Tjia, David, and Hendrik (2025) focused on the development of a web-based inventory system using the Rapid Application Development (RAD) methodology. The primary objective of the study was to improve inventory management processes through automation and digitalization. The researchers designed the system to enable users to submit requests for inventory items electronically while allowing administrators to review and approve requests before releasing the requested items.

The findings revealed that the system significantly improved inventory management by providing real-time stock monitoring, automated record keeping, and a centralized platform for request processing and approval. The implementation of the system reduced delays in inventory transactions, minimized human errors associated with manual record keeping, and improved the accuracy and reliability of inventory data. The study also found that the system enhanced communication between users and administrators and generated reports that supported inventory tracking and decision-making.

Furthermore, the researchers observed that the use of the RAD methodology enabled rapid system development and continuous improvement based on user feedback. The system improved accountability and control over inventory resources by ensuring that all transactions were properly authorized and documented. In addition, security measures limited system access to authorized personnel, thereby protecting inventory records from unauthorized modifications.

The researchers concluded that the web-based inventory system improved operational efficiency, inventory accuracy, accountability, and resource management. The study also emphasized the importance of automated inventory monitoring in preventing stock shortages and overstocking situations. Future enhancements were recommended to further improve the scalability and effectiveness of the system.

This study is relevant to the present research because it highlights the importance of web-based inventory monitoring, automated request processing, approval workflows, and report generation in improving business operations. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital inventory systems can improve efficiency, reduce errors, and strengthen inventory control.

The study conducted by Rakhman, Bakhar, and Afriliana (2025) focused on the development of a web-based inventory management system with a request and approval workflow. The purpose of the study was to improve inventory control and overcome the limitations of manual inventory management practices. The researchers designed the system to allow users to submit inventory requests electronically while administrators review and approve requests before releasing inventory items.

The findings revealed that the system significantly reduced paperwork and administrative workload by digitizing inventory transactions and records. The approval workflow ensured that inventory withdrawals were properly authorized, thereby reducing the risk of unauthorized use of inventory resources. The study also found that digital record keeping improved data accuracy, accessibility, and transparency in inventory operations.

Furthermore, the system maintained detailed transaction histories that enabled administrators to monitor inventory movement and stock levels in real time. The researchers observed that the system improved accountability among users and administrators because every transaction was properly documented and traceable. The availability of organized and updated inventory data also supported faster and more effective decision-making processes.

The researchers concluded that the inventory management system enhanced operational efficiency, inventory security, and resource management. The study also noted that the system improved user satisfaction by providing a faster and more convenient inventory request process. Continuous system improvement was recommended to accommodate future inventory management requirements and additional functionalities.

This study is relevant to the present research because it emphasizes the value of request-and-approval workflows, inventory monitoring, transaction tracking, and digital

record management. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how automated inventory systems can improve accountability, operational efficiency, and inventory security.

The study conducted by Odasco and Saong (2023) examined the implementation of an inventory management system designed to improve the management of non-consumable resources within a university environment. The system was developed to facilitate inventory monitoring, request processing, approval procedures, and transaction recording through an electronic platform.

The findings revealed that the system improved the accuracy of inventory records compared to traditional manual methods. Users were able to submit requests electronically, while administrators reviewed and approved transactions before issuing inventory items. The automated monitoring feature provided real-time updates on inventory availability and status, allowing administrators to manage inventory resources more efficiently.

Furthermore, the study found that the system maintained detailed transaction records that enhanced transparency and accountability. The borrowing and return processes became faster and more organized, reducing delays and improving service delivery. Accurate and updated inventory information also supported better planning, resource allocation, and decision-making within the institution.

The researchers concluded that the inventory management system strengthened inventory control, improved operational efficiency, and enhanced service quality. The study also emphasized the importance of maintaining accurate inventory records and properly documented transactions to ensure responsible resource utilization. Further system development was recommended to address the growing inventory management needs of the institution.

This study is relevant to the present research because it demonstrates how automated inventory monitoring, request processing, approval workflows, and transaction tracking can improve operational efficiency and resource management. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights the benefits of digital inventory systems in maintaining accurate records, improving accountability, and supporting informed decision-making.

The study conducted by Kumar and Patel (2020) examined the implementation of automated request-and-approval mechanisms in enterprise inventory management systems among manufacturing and retail firms in the United Kingdom. The researchers aimed to evaluate how structured approval workflows could improve inventory accuracy, accountability, and operational efficiency. The study was conducted in response to inventory discrepancies, inaccurate records, and unauthorized withdrawals caused by the lack of formal inventory authorization procedures.

The findings revealed that the implementation of automated request-and-approval workflows significantly reduced unauthorized inventory access and improved inventory tracking. Users were required to submit formal inventory requests containing item details, quantities, and purposes before inventory could be released. Designated personnel reviewed and approved requests based on inventory availability and organizational requirements.

Furthermore, the study found that the system generated complete audit trails for all inventory transactions, making it easier for administrators to monitor inventory movement and conduct internal audits. Managers were able to make better inventory decisions by reviewing stock levels before approving requests. The researchers also observed that employees gradually adapted to the approval process and recognized its role in promoting accountability and transparency within the organization.

The researchers concluded that automated request-and-approval mechanisms are essential features of modern inventory management systems because they strengthen inventory control, improve resource utilization, reduce inventory-related errors, and support effective

organizational decision-making. The study emphasized that structured approval workflows contribute significantly to inventory security and operational efficiency.

This study is relevant to the present research because it highlights the importance of inventory request approval, transaction monitoring, audit trails, and accountability in inventory management. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how approval-based inventory systems can improve inventory control, transparency, and resource management.

The study conducted by Zhang, Chen, and Liu (2021) investigated the adoption of digital inventory control systems with hierarchical approval workflows in manufacturing enterprises in China. The researchers aimed to determine how approval-based inventory systems could improve inventory accuracy, reduce material waste, and enhance supply chain performance. The study addressed problems associated with manual inventory procedures, including inaccurate inventory records, overstocking, shortages of essential materials, and inefficient procurement planning.

The findings revealed that the implementation of hierarchical approval workflows significantly improved inventory accuracy and reduced unnecessary inventory consumption. Inventory requests were required to pass through different levels of approval depending on the value and importance of the requested items. Supervisors and managers reviewed requests using real-time inventory information before granting approval.

Furthermore, the system incorporated automated notifications and monitoring features that ensured inventory requests were processed within established timeframes. The researchers found that properly documented inventory withdrawals improved accountability and transparency while reducing material waste. The collected inventory request data also provided valuable information for demand forecasting and procurement planning.

The researchers concluded that hierarchical approval systems play an important role in strengthening inventory control, promoting accountability, reducing unnecessary inventory consumption, and improving operational efficiency. The study emphasized that integrating approval mechanisms into digital inventory systems contributes to better organizational performance and resource management.

This study is relevant to the present research because it demonstrates the effectiveness of approval-based inventory control, automated monitoring, and inventory tracking in improving inventory management processes. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights how digital inventory systems can improve inventory accuracy, accountability, and decision-making while ensuring proper authorization of inventory transactions.

BUSINESS INTELLIGENCE

The adoption of information systems has become increasingly important for small and medium-sized enterprises (SMEs) seeking to improve operational efficiency and maintain competitiveness. According to Salisu et al. (2021), the successful adoption of business information systems is influenced by technological, organizational, environmental, and managerial factors. The study utilized the Technology-Organization-Environment (TOE) Framework and the Information Systems Adoption Model for Small Businesses to identify the key determinants affecting the implementation of information systems within SMEs.

The researchers identified several technological factors that influence system adoption, including relative advantage, compatibility, complexity, trialability, and observability. These factors determine how organizations perceive the benefits and usability of a new system. The study emphasized that businesses are more likely to adopt technological innovations when they provide clear advantages, align with existing business processes, and demonstrate practical value in improving organizational performance.

Furthermore, organizational factors such as management support, organizational size, resource availability, absorptive capacity, and the presence of innovation champions were found to play a significant role in successful system implementation. The study highlighted that strong managerial support and adequate organizational resources are essential in facilitating technology adoption and ensuring the effective utilization of information systems within the organization.

The study also examined environmental factors, including vendor support, competitive pressure, and government support. The findings revealed that external pressures and support

mechanisms encourage businesses to adopt information systems to enhance their competitiveness and operational capabilities. In addition, owner-managers' characteristics, such as innovativeness, information technology knowledge, and positive attitudes toward technology, were identified as critical factors influencing the decision to implement new systems.

The findings of the study are relevant to the present research because they emphasize the importance of organizational readiness, management support, technological compatibility, and user acceptance in the successful implementation of information systems. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates that adopting an integrated information system can improve business operations, support decision-making, and enhance overall organizational efficiency. The identified factors may serve as valuable considerations in ensuring the successful implementation and acceptance of the proposed system within the business.

Mavutha et al. (2023) examined the factors influencing the adoption of Business Intelligence (BI) systems among micro-small apparel retailers in KwaZulu-Natal, South Africa. The study aimed to identify the technological, organizational, and environmental factors that affect the adoption of Business Intelligence systems within small businesses. Using the Technology-Organization-Environment (TOE) Framework, the researchers conducted a quantitative cross-sectional survey involving 132 apparel business owners.

The findings revealed that perceived relative advantage, organizational readiness, management support, and organizational data environment positively influence the adoption of Business Intelligence systems. The study emphasized that business owners are more likely to adopt information technologies when they perceive clear benefits, have adequate organizational resources, and receive strong managerial support. Furthermore, the researchers found that information systems contribute to improved decision-making, operational efficiency, and business competitiveness.

The study also identified cost as one of the primary barriers to technology adoption among small businesses. Despite this challenge, many respondents recognized the long-term benefits of adopting information systems and expressed willingness to invest in technological solutions that could improve business performance. The findings suggest that successful implementation of information systems requires not only technological readiness but also organizational commitment and strategic planning.

The study is relevant to the present research because it highlights the importance of adopting information systems within small business environments. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how information technology can enhance operational efficiency, support

informed decision-making, and improve overall business performance. The identified factors influencing technology adoption may also serve as valuable considerations in ensuring the successful implementation and acceptance of the proposed system.

Chen (2025) conducted a study entitled *Applications of Artificial Intelligence in Small and Medium Scale Business* to investigate the adoption, implementation, and impact of Artificial Intelligence (AI) among Small and Medium-Sized Enterprises (SMEs) across 12 countries in North America, Europe, and Asia-Pacific. The study utilized a mixed-methods research design involving 583 survey respondents, 47 semi-structured interviews, and 18 case studies to examine how SMEs use AI technologies to improve business operations and competitiveness.

The findings revealed that 64.7% of SMEs had implemented at least one AI application, particularly in customer service, marketing, and operational functions. The study found that AI implementation significantly improved operational efficiency by 27.3%, customer satisfaction by 24.8%, and cost reduction by 22.4%. Furthermore, the research identified key success factors for technology adoption, including clear problem identification, leadership commitment, data quality, workflow integration, and user training. The study also reported that businesses utilizing digital technologies achieved faster decision-making, increased productivity, and improved overall business performance.

The researchers concluded that technology-driven systems play a significant role in enhancing business operations, improving efficiency, and supporting organizational growth. The study emphasized that the successful implementation of digital solutions enables businesses to streamline processes, manage resources effectively, and make informed decisions.

This study is relevant to the present research because it demonstrates how technology-based systems can improve operational efficiency, data management, and overall business performance. Similarly, the proposed Web-Based Sales, Attendance, and Inventory

Management System for Mr. Softy aims to automate manual processes, improve inventory monitoring, manage employee attendance efficiently, and provide accurate sales records to support better decision-making and business operations.

Local Studies

POINT OF SALES

The study conducted by the researchers examined the utilization of Point-of-Sale (POS) systems among selected start-up cafés in Tabaco City, Albay. Using a qualitative descriptive research design, the study gathered data from café owners, managers, and staff to determine the features, benefits, challenges, and utilization of POS systems in daily business operations.

The findings revealed that POS systems were primarily used for basic transaction functions such as order processing, billing, receipt generation, and sales recording. These features helped improve transaction accuracy, reduce waiting time, automate reporting, and support real-time monitoring of business activities. The study also found that inventory management, sales analytics, and customer relationship management features were available but were not fully utilized by many café operators due to limited technical knowledge, inadequate training, and operational constraints.

Furthermore, the study identified several benefits of POS systems, including improved operational efficiency, reduced human errors, enhanced customer service, better sales monitoring, and increased transparency in transactions. However, challenges such as unstable internet connectivity, system downtime, maintenance costs, limited integration capabilities, and insufficient employee training affected the effective utilization of advanced POS features.

The researchers concluded that while POS systems significantly improve business operations, many small enterprises still utilize only the basic functions of the technology. The study recommended providing additional training, improving technological infrastructure,

and maximizing advanced POS features to enhance business competitiveness and operational sustainability.

This study is relevant to the present research because it highlights the importance of automated sales processing, inventory monitoring, reporting, and employee management in improving business efficiency. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how integrated digital systems can reduce manual work, improve record accuracy, and support better decision-making for small business operations.

Divina et al. (2025) developed a web-based Point-of-Sale (POS) and Inventory Management System for a small food enterprise known as “Pastil sa Tabi.” The study aimed to automate sales transactions, inventory monitoring, and forecasting processes to address inefficiencies associated with manual record-keeping. The system was developed using PHP, MySQL, HTML, CSS, and JavaScript following the Waterfall Software Development Life Cycle (SDLC) model. It integrated transaction processing, inventory management, sales forecasting, and predictive analytics to improve operational efficiency and support data-driven decision-making.

The researchers evaluated the system using the ISO/IEC 25010 software quality standards with fifty (50) respondents composed of end-users and technical experts. The results showed that the system received high ratings in terms of functionality, usability, reliability, effectiveness, and security. The implementation of real-time inventory monitoring, sales reporting, and forecasting features enabled the business to improve inventory control, reduce shortages and waste, and make informed decisions regarding stock replenishment and resource allocation.

Furthermore, the study concluded that integrating POS, inventory management, and predictive analytics into a single platform significantly enhanced business productivity, operational accuracy, and customer service. The researchers emphasized that adopting automated systems can help small businesses improve efficiency, sustainability, and profitability through better management of sales and inventory data.

This study is relevant to the present research because it demonstrates how a web-based POS and Inventory Management System can streamline business operations and improve data management. Similar to the proposed Sales, Attendance, and Inventory Management

System for Mr. Softy, the study highlights the importance of automation in enhancing operational efficiency, maintaining accurate records, and supporting effective decision-making. The findings serve as a basis for developing a system that integrates sales monitoring, inventory control, and employee management into a centralized platform.

Moreno et al. (2025) examined the challenges and perceptions of micro and small enterprises in Tacurong City regarding the adoption and use of Point of Sale (POS) systems. The study found that most businesses utilized legacy POS systems and generally experienced only minimal challenges in operating the technology. However, operational issues such as power interruptions, network reliability, and maintenance costs were identified as common concerns among users.

The findings revealed that business owners and managers had a positive perception of POS systems, particularly in terms of perceived usefulness and ease of use. Respondents believed that POS systems improved business performance, increased productivity, enhanced operational efficiency, and simplified daily business activities. The study further showed that operational challenges significantly influenced users' perceptions of POS systems, indicating that fewer operational difficulties resulted in a more favorable attitude toward system adoption.

The study is relevant to the present research because it highlights the importance of computerized systems in improving business operations. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital technologies can enhance efficiency, streamline transactions, improve inventory monitoring, and support business decision-making. The findings provide evidence that technology-based management systems contribute to better organizational performance and can encourage businesses to adopt more advanced and integrated solutions.

According to Mangmang (2018), Point-of-Sale (POS) systems serve as effective tools in managing sales transactions, inventory monitoring, and business profitability within retail establishments. The study entitled “Development and Implementation of Point of Sale System (POS): Profitability Measurement for Retail Business” was conducted to develop and implement an enhanced POS system with notification features for overdue payments and out-of-stock items. The primary objective of the study was to improve business operations through automation and better monitoring of retail transactions.

The researcher utilized a developmental-evaluative research design and developed the system using Microsoft Visual Studio 2012 and Microsoft SQL Server 2005. The software was evaluated based on the ISO/IEC 25010:2011 software quality standards, specifically focusing on effectiveness, usability, satisfaction, and performance efficiency. The owner and cashier of the retail business served as respondents in evaluating both the existing and developed systems.

The findings revealed that the developed POS system significantly improved business operations by providing efficient monitoring of inventory levels, overdue customer payments, and stock availability. The implementation of notification features enabled users to track out-of-stock items and overdue accounts more effectively, reducing manual monitoring efforts and improving decision-making. The results further showed that the developed system achieved higher levels of satisfaction, effectiveness, and performance efficiency compared to the existing manual system.

The study concluded that adopting technological innovations such as POS systems can greatly improve retail business operations, increase efficiency, enhance inventory management, and support profitability. Furthermore, the researcher emphasized that

businesses handling credit transactions can benefit from automated monitoring features that help ensure timely payments and better financial management.

This study is relevant to the present research because it supports the development of a Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy. Similar to the proposed system, the study highlights the importance of automating sales transactions and inventory monitoring to improve operational efficiency and business performance. The findings demonstrate how computerized systems can provide accurate records, efficient monitoring, and reliable information that support better business management and decision-making.

According to Canlas, Collado, Magundayao, Sy, and Tumbali (2021), Point-of-Sale (POS) systems significantly influence the supply chain performance of retail businesses by improving inventory monitoring, transaction processing, customer service, and business reporting. The study was conducted among selected retail establishments in Tuguegarao City, Cagayan, to examine the influence of POS technologies on supply chain performance. Using a descriptive-correlational research design, the researchers gathered data from employees working in procurement, inventory management, finance, information technology, and marketing departments of retail businesses utilizing POS systems.

The study evaluated three major POS components: Rapid Scan Systems, Cloud-Based Communication Systems, and Electronic Funds Transfer at Point of Sale (EFTPOS). The findings revealed that Rapid Scan Systems improved transaction speed, reduced customer waiting time, minimized data processing errors, and enhanced inventory tracking. Cloud-Based Communication Systems enabled businesses to generate performance reports, monitor inventory levels, identify purchasing patterns, and support data-driven decision-making. Meanwhile, EFTPOS provided secure and convenient cashless transactions that improved customer convenience and operational efficiency.

The results further showed that POS systems contributed to high supply chain performance by reducing operational costs, increasing customer satisfaction, improving service speed, enhancing supplier relationships, increasing sales, and strengthening competitive advantage. The study also found that cloud-based communication systems had a significant relationship with supply chain responsiveness, service speed, and customer satisfaction, demonstrating the importance of real-time information access in business operations.

The researchers concluded that POS systems play a vital role in improving retail business performance through automation, real-time data processing, efficient inventory management, and enhanced customer service. The adoption of POS technologies allows businesses to streamline operations, improve decision-making, and increase overall organizational efficiency.

This study is relevant to the present research because it supports the development of a Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy. Similar to the proposed system, the study highlights the importance of integrating sales transactions, inventory monitoring, and business reporting into a computerized platform. The findings demonstrate how automation and real-time information management can improve operational efficiency, support informed decision-making, and enhance overall business performance.

ATTENDANCE MONITORING

The study conducted by Domingo and Ladia (2024) focused on the development of QSUM-eASys, a Face Recognition Attendance and Web-Based Attendance Monitoring System for Quirino State University Maddela Campus. The purpose of the study was to address the limitations of the institution's existing biometric attendance device, which only generated raw attendance summaries that were insufficient for payroll processing and formal record keeping. The researchers developed a web-based system that integrated attendance data from a facial recognition device and transformed it into a centralized and reportable format for attendance processing officers.

The findings revealed that the system effectively improved attendance monitoring by providing accurate tracking of employee time-in and time-out records. The web-based platform centralized attendance information, allowing authorized personnel to access, monitor, and manage attendance records efficiently. The researchers found that the system streamlined attendance processing and provided reliable reports that could be directly utilized for payroll registry preparation during each payroll cutoff period.

Furthermore, the system enhanced security and accountability by incorporating measures that prevented fraudulent attendance recording and unauthorized manipulation of attendance records. The researchers observed that the system improved administrative efficiency and reduced the workload associated with manual attendance consolidation and report preparation.

The researchers concluded that the face recognition attendance and monitoring system provides an effective solution for employee attendance management, payroll support, and

attendance record monitoring. The study emphasized the importance of integrating attendance technologies with web-based reporting systems to improve operational efficiency and data reliability.

This study is relevant to the present research because it highlights employee attendance monitoring, facial recognition technology, web-based record management, and payroll integration. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how automated attendance systems can improve accuracy, accountability, operational efficiency, and employee management.

According to Calo et al. (2021), the development of an In-Classroom Faculty Attendance Monitoring System using Ultra High Frequency (UHF) Radio Frequency Identification (RFID) with image cross-verification significantly improved the efficiency and reliability of attendance monitoring. The study was conducted to address the limitations of traditional attendance monitoring systems that only record daily time-in and time-out logs and are unable to verify actual attendance within the workplace.

The researchers developed a system composed of a UHF-RFID reader, camera module, Raspberry Pi device, and a web-based attendance monitoring application connected to a MySQL database. The system automatically recorded attendance by detecting RFID-enabled identification cards within a two-meter range and generated attendance reports through a web-based platform. Additionally, image capture was utilized as a verification mechanism to minimize fraudulent attendance practices.

The findings revealed that the system provided accurate attendance monitoring, improved data reliability, reduced manual monitoring tasks, and received positive evaluations in terms of functionality and usability. The study concluded that automated attendance monitoring systems can enhance attendance management efficiency while providing dependable and real-time attendance information.

This study is relevant to the present study because it demonstrates how automated attendance monitoring technologies can improve attendance recording, monitoring, and reporting processes. The findings support the integration of attendance monitoring features in business management systems to promote accountability, operational efficiency, and accurate employee attendance tracking.

According to Reynoso and Torres (2020), the development of Tempus, a facial recognition technology-based attendance monitoring system, significantly improved the efficiency and accuracy of attendance management. The study aimed to replace traditional attendance monitoring methods by integrating facial recognition technology, Internet of Things (IoT) capabilities, and mobile platform support into a single automated system.

The researchers developed Tempus using Raspberry Pi 3 as the primary hardware component, Haar Cascade for facial detection, and Local Binary Pattern Histogram (LBPH) for facial recognition. The system was designed to record attendance automatically, manage schedules, generate reports, and provide real-time notifications to users through a mobile application. In addition, the system allowed administrators to monitor attendance records, manage user accounts, and customize attendance settings according to organizational requirements.

The findings revealed that the system achieved excellent ratings in terms of functionality, usability, accuracy, efficiency, and reliability. The integrated computer vision technology, real-time attendance monitoring, mobile support, and user-friendly interface contributed to the effectiveness of the system. Furthermore, the facial recognition component demonstrated an average accuracy rate of 83 percent in real-time attendance identification.

The study concluded that automated attendance monitoring systems enhance attendance management by providing accurate attendance records, real-time reporting, and efficient information dissemination. The integration of smart technologies also improves monitoring processes and supports decision-making within organizations.

This study is relevant to the present study because it demonstrates how automated attendance monitoring systems can improve employee attendance tracking, report generation,

and attendance management. Although the present study does not utilize facial recognition technology, the findings support the implementation of attendance monitoring features to promote operational efficiency, accountability, and accurate attendance recording in business establishments such as Mr. Softy.

According to Fernandez et al. (2025), the development of a Facial and Geofencing-Based Attendance Tracking System significantly improved attendance monitoring, personnel accountability, and record accuracy among deployed personnel. The study was conducted for the Cagayan de Oro City Police Office (COCPO), Philippines, to address issues associated with manual attendance recording, inaccurate records, and proxy attendance practices.

The researchers developed a mobile and web-based attendance tracking system that integrated facial recognition and geofencing technologies. The system required personnel to verify their identity through facial recognition and confirm their location within a designated geofence before attendance could be recorded. Attendance information was then transmitted to a centralized database where administrators could monitor records and generate reports through a web dashboard.

The findings revealed that the system achieved a facial recognition accuracy of 99 percent under optimal conditions and a geofencing accuracy of 92.18 percent in open environments. Furthermore, the system obtained a System Usability Scale (SUS) score of 85.56, indicating a high level of usability and user acceptance. The researchers concluded that the system enhanced transparency, accountability, and efficiency while reducing errors associated with traditional attendance monitoring methods.

This study is relevant to the present study because it demonstrates the effectiveness of automated attendance monitoring systems in improving attendance recording, monitoring, and report generation. Although the proposed system for Mr. Softy does not utilize facial recognition and geofencing technologies, the findings support the implementation of computerized attendance monitoring features to ensure accurate employee attendance tracking, improve operational efficiency, and strengthen workforce accountability.

INVENTORY MANAGEMENT

The study conducted by Cortez, Meneses, and Tan Jr. (2024) focused on the design and development of a Laboratory Inventory Management System (LIMS) for educational institutions in the Philippines. The researchers aimed to improve the management of laboratory equipment, materials, and supplies through a computerized inventory system. The study recognized that manual inventory processes often result in inaccurate records, misplaced items, and delays in inventory monitoring.

The findings revealed that the system improved inventory management by providing sample management, instrument documentation access, and centralized data storage. Users were able to record, update, and retrieve inventory information efficiently through a computerized database. The system also enabled laboratory personnel to monitor inventory levels and equipment status more effectively.

Furthermore, the researchers found that the implementation of the system improved inventory accuracy and significantly reduced manual errors. The automated process shortened the time required to locate inventory information and generate reports. The study also revealed that all inventory transactions were properly documented, resulting in greater accountability and transparency. Real-time monitoring supported informed decision-making, resource planning, and proper allocation of laboratory assets.

The researchers concluded that the implementation of the Laboratory Inventory Management System significantly enhanced inventory accuracy, operational efficiency, accountability, and overall inventory management. The study emphasized that computerized

inventory systems are effective tools for maintaining organized records and improving resource utilization.

This study is relevant to the present research because it demonstrates the importance of automated inventory monitoring, centralized record management, and real-time reporting in improving organizational efficiency. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights how digital inventory systems can reduce errors, improve record accuracy, and support better business decision-making.

The study conducted by Anade, Calibo, Noveda, Yu, and Verecio (2023) developed a Supply Inventory Management System for Leyte Normal University. The researchers sought to address common problems associated with manual inventory management, including inaccurate records, delays in updating information, and difficulties in monitoring inventory movement.

The findings revealed that the system improved inventory monitoring and reporting through the use of a centralized database accessible to authorized personnel. Users were able to record transactions, update inventory information, and generate reports efficiently. The implementation of automation improved the accuracy and reliability of inventory records while reducing paperwork and minimizing human errors.

Furthermore, the study found that real-time monitoring enabled administrators to track inventory levels and movement more effectively. Digital transaction recording improved transparency and accountability because inventory activities became traceable and easier to audit. The system also helped identify inventory shortages and supported inventory replenishment planning. Users reported that the computerized system was more convenient and efficient than traditional manual methods.

The researchers concluded that the system enhanced inventory control, operational efficiency, transparency, and resource management. The study also emphasized the importance of technology adoption in modern inventory management practices and highlighted the role of accurate inventory data in supporting effective decision-making.

This study is relevant to the present research because it demonstrates how automated inventory monitoring, transaction recording, and report generation can improve business operations. Similar to the proposed Web-Based Sales, Attendance, and Inventory

Management System for Mr. Softy, the study shows how digital inventory solutions can improve inventory accuracy, operational efficiency, and resource management.

The study conducted by Dela Cruz and Santos (2021) focused on the development of a web-based inventory management system for a state university in Laguna, Philippines. The researchers aimed to improve inventory control and address problems associated with manual inventory processes, particularly inventory discrepancies caused by undocumented withdrawals and inaccurate stock records.

The findings revealed that the system successfully implemented a request-and-approval workflow that required users to submit item requests before supplies could be released. Inventory officers reviewed and approved requests based on stock availability, while all transactions were digitally recorded for monitoring and auditing purposes. The system also provided real-time inventory tracking, enabling administrators to monitor inventory movement and stock levels more effectively.

Furthermore, the study found that the implementation of the system resulted in a significant reduction in inventory discrepancies and improved accountability among users. Digital records simplified inventory tracking, supported procurement planning, and improved transparency within inventory operations. The researchers observed that the request-and-approval process strengthened inventory control by ensuring that all inventory transactions were properly authorized and documented.

The researchers concluded that integrating a request-and-approval workflow into an inventory management system improves transparency, inventory accuracy, accountability, and resource management. The study emphasized the value of digital inventory monitoring in maintaining accurate inventory records and supporting effective operational management.

This study is highly relevant to the present research because it highlights the importance of web-based inventory monitoring, approval workflows, and transaction tracking. Similar to

the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how automated inventory processes can improve inventory control, accountability, and operational efficiency.

The study conducted by Odasco and Saong (2023) analyzed the University of Baguio Requisition, Procurement, and Inventory System (UBRPIS) and its effectiveness in supporting inventory management operations. The researchers aimed to evaluate the system's contribution to inventory control and service delivery within the university. The study examined employees' familiarity with the system and identified its strengths and limitations in managing inventory resources.

The findings revealed that the automated system streamlined inventory recording, monitoring, and updating of records. Employees reported that the system made inventory-related tasks more efficient, organized, and easier to manage. The researchers found that automation reduced delays commonly associated with manual inventory processes while improving the accuracy and accessibility of inventory information.

Furthermore, the study showed that the system enabled administrators to monitor inventory movement and stock availability more effectively. The integration of barcode technology improved inventory tracking and monitoring, while properly documented transactions increased accountability and transparency. The researchers also identified operational challenges and recommended additional system enhancements and user training to maximize system utilization.

The researchers concluded that automated inventory systems play an important role in improving efficiency, transparency, accountability, and inventory control. The study emphasized that effective inventory management contributes to better service delivery, informed decision-making, and proper resource allocation within organizations.

This study is relevant to the present research because it demonstrates the effectiveness of automated inventory monitoring, transaction recording, and inventory tracking technologies

in improving organizational operations. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study highlights how digital inventory systems can improve inventory accuracy, operational efficiency, and resource management.

The study conducted by Reyes, Fontanilla, and Magno (2020) examined the inventory and procurement practices of selected local government units (LGUs) in the Visayas region of the Philippines. The researchers identified several challenges associated with manual inventory and procurement processes, including delays in request processing, misplaced documents, weak request tracking, inconsistent record-keeping, and limited visibility regarding the status of submitted requests.

The findings revealed that the computerized inventory and procurement management system successfully digitized the request-and-approval workflow. Authorized users were able to submit supply requests electronically, while approving officers reviewed requests through a centralized dashboard containing real-time inventory information. The system also provided automatic notifications that reduced the need for manual follow-ups and improved communication between users and administrators.

Furthermore, the study found that digital transaction recording improved inventory monitoring, reporting, transparency, and accountability. Administrators were able to track inventory movement more effectively and generate reports for auditing and decision-making purposes. The pilot implementation demonstrated a significant reduction in processing time, with requests being completed in less than 24 hours compared to several days under the manual process. Inventory accuracy and operational efficiency also improved due to better documentation and tracking mechanisms.

The researchers concluded that computerized request-and-approval systems enhance efficiency, strengthen governance, reduce operational delays, and improve inventory and procurement management. The study emphasized the importance of digital systems in maintaining accurate records and supporting transparent resource management.

This study is highly relevant to the present research because it highlights the benefits of request-and-approval workflows, inventory monitoring, transaction tracking, and automated reporting. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the study demonstrates how digital inventory systems can improve operational efficiency, accountability, transparency, and inventory control.

BUSINESS INTELLIGENCE

The study conducted by Paulino (2022) examined the influence of Business Intelligence (BI) generated through Business Analytics (BA) on organizational performance within the Philippine retail industry. Using Partial Least Squares Structural Equation Modeling (PLS-SEM), the study analyzed the relationship between business analytics capabilities, business intelligence, and organizational performance among retail organizations.

The findings revealed that business analytics significantly contributes to the generation of business intelligence, which in turn enhances organizational performance. The study showed that BI improves marketing performance, financial performance, and business process efficiency by providing managers with accurate, timely, and relevant information for decision-making. Furthermore, the research emphasized that organizations utilizing business intelligence tools are more capable of monitoring business operations, identifying trends, and implementing data-driven strategies.

The study concluded that business intelligence serves as an essential resource for organizations seeking to improve competitiveness and operational effectiveness. Through the use of analytical tools and information systems, businesses can gain valuable insights that support strategic planning and organizational growth.

This study is relevant to the present research because it highlights the importance of utilizing computerized systems to collect, process, and analyze business data. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, business intelligence capabilities can provide accurate reports, improve monitoring processes, and support informed managerial decision-making.

The study conducted by Reginio et al. (2024) focused on the development and evaluation of a Community Affairs Management Information System (CAMIS) for the Municipality of Torrijos, Marinduque. The researchers designed the system to improve data management, record keeping, reporting processes, and information accessibility for community affairs operations. Using a developmental research approach, the study assessed the system's effectiveness in managing records and supporting administrative functions.

The findings revealed that the implementation of the management information system significantly improved the organization, storage, retrieval, and processing of information. The system enabled administrators to generate reports more efficiently, access records quickly, and monitor community programs using real-time data. The researchers also found that the system enhanced data accuracy, reduced manual processing errors, and improved overall operational efficiency.

Furthermore, the study emphasized that integrating analytical and reporting functions into information systems allows organizations to make informed decisions based on accurate and updated information. The availability of organized records and automated reports contributed to better planning, monitoring, and evaluation of organizational activities.

The researchers concluded that management information systems play a vital role in improving organizational performance through efficient information management and decision support. The study recommended the adoption of computerized information systems to enhance productivity, transparency, and service delivery.

This study is relevant to the present research because it demonstrates how a web-based management system can improve data organization, reporting accuracy, and operational efficiency. Similar to the proposed Web-Based Sales, Attendance, and Inventory Management

System for Mr. Softy, the use of automated information management and reporting tools can streamline business processes, reduce manual errors, and support effective managerial decision-making.

The study conducted by Silaya and Baldovino (2025) examined the relationship between business analytics and organizational functions among small and medium enterprises (SMEs) in Quezon Province, Philippines. The researchers investigated how descriptive, predictive, and prescriptive analytics contribute to business planning, operational management, performance monitoring, and strategic decision-making. Using a quantitative research approach, the study evaluated the extent to which analytics practices support organizational effectiveness.

The findings revealed that business analytics significantly improves organizational functions by providing valuable insights derived from business data. The study showed that analytics tools assist managers in monitoring business performance, identifying trends, forecasting future outcomes, and making evidence-based decisions. The researchers found that organizations that utilize business analytics are more capable of improving operational efficiency, resource allocation, and strategic planning.

Additionally, the study emphasized that data-driven decision-making enhances the ability of SMEs to respond to changing business environments and maintain competitiveness. The use of analytics technologies enabled organizations to transform raw business data into meaningful information that supports organizational growth and sustainability.

The researchers concluded that business analytics serves as an important component of modern business management. The study recommended that SMEs adopt analytics tools and information systems to improve decision-making processes and strengthen overall business performance.

This study is relevant to the present research because it highlights the importance of collecting, analyzing, and utilizing business data to support managerial decisions. Similar to

the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy, the integration of reporting and analytics features can help monitor sales, inventory levels, and employee attendance while providing useful information for improving business operations and productivity.

Related Literature

Point of Sale (POS) System

Point-of-Sale (POS) systems have become one of the most important technologies used by businesses to improve operational efficiency and accuracy in transaction processing. A POS system is a combination of hardware and software that enables businesses to process sales transactions, monitor inventory, generate reports, and manage customer information. Unlike traditional manual transaction methods, POS systems automate business processes, reducing the possibility of human errors and improving the speed of service delivery.

According to Asrani et al. (2024), POS systems integrate various business functions such as sales processing, inventory management, payment handling, and report generation into a single platform. Their study emphasized that web-based POS systems provide real-time access to business data, allowing managers to monitor sales performance and inventory levels from any location. This capability supports faster and more informed decision-making, which is essential in maintaining business competitiveness.

Similarly, Rahangdale et al. (2024) developed a POS Inventory Management System to address the inefficiencies of manual record-keeping. Their findings revealed that the computerized system improved inventory tracking, reduced data-entry errors, and enhanced overall business productivity. The researchers emphasized that automation enables businesses to maintain accurate inventory records and generate reliable reports that support operational planning.

Ramadhani, Nindyasari, and Murti (2025) further highlighted the effectiveness of integrating sales and inventory management within a web-based POS platform. Their study demonstrated that automation significantly improved inventory accuracy, reduced operational delays, and streamlined business processes. The integration of multiple business functions into a centralized system allowed managers to monitor business activities efficiently and make timely decisions.

These studies collectively indicate that POS systems are valuable tools for improving transaction accuracy, inventory control, and overall business performance. Their findings support the proposed Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy because the sales module of the proposed system aims to automate transactions, maintain accurate sales records, and provide management with reliable reports for decision-making.

Attendance Monitoring System

Attendance monitoring plays a significant role in workforce management because it directly affects employee productivity, accountability, and payroll computation. Traditional attendance methods such as manual logbooks and attendance sheets often result in inaccurate records, delayed reporting, and opportunities for attendance fraud. As a result, many organizations have adopted computerized attendance monitoring systems to improve efficiency and reliability.

Mohamad Noor, Ibrahim, and Jamaluddin (2023) developed an Employee Attendance Management System using Flutter technology to provide a digital solution for attendance recording. Their study revealed that the system improved attendance accuracy, reduced administrative workload, and enabled real-time attendance monitoring through a centralized database. The researchers concluded that mobile-based attendance systems provide organizations with a more convenient and reliable method of managing employee attendance.

Likewise, Nazara and Nasien (2024) developed an attendance system integrated with Location-Based Service (LBS) technology. Their findings showed that GPS verification significantly improved attendance validation by ensuring that employees could only record attendance within authorized locations. This feature reduced attendance fraud and strengthened employee accountability.

On the local level, Domingo and Ladia (2024) developed QSUM-eASys, a web-based attendance monitoring system integrated with facial recognition technology. The system effectively tracked employee attendance, generated payroll-ready reports, and improved administrative efficiency through centralized attendance management. Similarly, Reynoso

and Torres (2020) developed Tempus, a facial recognition-based attendance system integrated with Internet of Things (IoT) technology. The system achieved high ratings in terms of functionality, usability, and reliability, proving the effectiveness of automated attendance monitoring solutions.

These studies demonstrate that attendance monitoring systems contribute significantly to improving attendance accuracy, reducing administrative tasks, and enhancing employee accountability. For the proposed Mr. Softy system, the attendance monitoring module will help management track employee attendance efficiently while maintaining accurate attendance records and supporting workforce management.

Inventory Management System

Inventory management is one of the most critical functions in business operations because it ensures that products are available when needed while preventing stock shortages and overstocking. Poor inventory management can lead to financial losses, operational delays, and reduced customer satisfaction. For this reason, businesses increasingly rely on computerized inventory management systems to improve inventory control and resource utilization.

Tjia, David, and Hendrik (2025) developed a web-based inventory management system using the Rapid Application Development (RAD) methodology. Their study found that the system improved inventory monitoring, automated record-keeping, and enhanced inventory control through real-time stock updates. The researchers emphasized that computerized inventory systems help organizations maintain accurate records while reducing human errors associated with manual inventory management.

Similarly, Rakhman, Bakhar, and Afriliana (2025) developed an inventory management system with a request-and-approval workflow. Their findings showed that the system strengthened inventory control by ensuring that all inventory withdrawals were properly authorized. The researchers concluded that approval-based inventory systems improve accountability, transparency, and resource management.

Locally, Anade et al. (2023) developed a Supply Inventory Management System for Leyte Normal University. Their study revealed that automation improved inventory accuracy, transparency, and operational efficiency through real-time monitoring and report generation. Cortez, Meneses, and Tan Jr. (2024) also developed a Laboratory Inventory Management

System that enhanced inventory tracking, record accuracy, and resource management through centralized data storage and automated monitoring.

The reviewed studies demonstrate that inventory management systems are essential for maintaining accurate inventory records, monitoring stock movement, and supporting operational efficiency. These findings justify the inclusion of an inventory management module in the proposed Mr. Softy system to ensure effective inventory control and accurate stock monitoring.

Business Intelligence

Business Intelligence (BI) refers to the technologies and processes used to collect, analyze, and transform business data into meaningful information that supports decision-making. In today's business environment, organizations generate large amounts of data through sales transactions, inventory records, customer interactions, and employee activities. BI tools help organizations analyze this data and convert it into valuable insights that improve organizational performance.

Paulino (2022) examined the influence of Business Intelligence generated through Business Analytics on organizational performance in the Philippine retail industry. The study revealed that BI improves marketing performance, financial performance, and business process efficiency by providing accurate and timely information for decision-making. The researcher emphasized that organizations using BI tools are better equipped to identify trends, monitor performance, and implement data-driven strategies.

Similarly, Mavutha et al. (2023) investigated the adoption of Business Intelligence systems among small businesses and found that BI technologies contribute to improved operational efficiency, organizational performance, and business competitiveness. The study highlighted the importance of management support, organizational readiness, and technological capability in successfully implementing BI systems.

Furthermore, Silaya and Baldovino (2025) found that business analytics significantly improves organizational functions by helping managers monitor performance, identify trends, forecast outcomes, and make evidence-based decisions. Their study emphasized that

data-driven decision-making enables businesses to respond effectively to changing market conditions and maintain competitiveness.

These studies indicate that Business Intelligence plays an essential role in supporting managerial decision-making and improving business performance. In the proposed Mr. Softy system, reporting and analytics features will serve as Business Intelligence tools by providing managers with accurate sales, attendance, and inventory information needed for effective decision-making.

Synthesis of Reviewed Literature

The reviewed literature and studies provide substantial evidence that information systems have become essential tools for improving operational efficiency, data management, and decision-making in modern organizations. The concepts and studies related to Point-of-Sale (POS) Systems, Attendance Monitoring Systems, Inventory Management Systems, Business Intelligence, and Web-Based Information Systems collectively emphasize the importance of automation in addressing the limitations of traditional manual processes. These technologies enable organizations to streamline business operations, improve record accuracy, reduce administrative workload, and generate reliable information that supports managerial decision-making.

The reviewed studies on Point-of-Sale systems consistently demonstrated the effectiveness of computerized sales processing in improving business operations. Studies conducted by Asrani et al. (2024), Rahangdale et al. (2024), Ramadhani et al. (2025), and other local researchers revealed that POS systems improve transaction accuracy, inventory monitoring, financial reporting, and customer service. These systems provide real-time access to business information, allowing managers to monitor sales performance, track inventory movement, and make timely decisions. The findings suggest that integrating sales and inventory management functions into a centralized platform significantly enhances productivity, operational efficiency, and business performance.

Similarly, the reviewed studies on Attendance Monitoring Systems highlighted the advantages of automating employee attendance recording and monitoring processes. The studies conducted by Mohamad Noor et al. (2023), Nazara and Nasien (2024), Domingo and Ladia (2024), Reynoso and Torres (2020), and other researchers demonstrated that attendance

monitoring systems improve attendance accuracy, reduce fraudulent attendance practices, enhance workforce accountability, and simplify attendance management. The integration of technologies such as web-based applications, GPS verification, RFID, facial recognition, and IoT further strengthened attendance validation and monitoring capabilities. These studies indicate that automated attendance systems contribute significantly to efficient workforce management and organizational productivity.

The reviewed Inventory Management System studies also revealed the importance of automation in inventory control and resource management. The studies conducted by Tjia et al. (2025), Rakhman et al. (2025), Anade et al. (2023), Cortez et al. (2024), and other researchers showed that computerized inventory systems improve inventory accuracy, minimize stock discrepancies, reduce losses, and support efficient monitoring of inventory movement. Real-time inventory updates, automated reporting, and approval-based workflows were identified as important features that enhance accountability and transparency. These findings demonstrate that inventory management systems play a crucial role in maintaining adequate stock levels, preventing inventory-related problems, and improving organizational efficiency.

Furthermore, the reviewed Business Intelligence studies emphasized the growing importance of data-driven decision-making in business operations. The studies of Paulino (2022), Mavutha et al. (2023), and Silaya and Baldovino (2025) revealed that Business Intelligence and Business Analytics enable organizations to transform raw data into meaningful information that supports planning, performance evaluation, forecasting, and strategic decision-making. The studies consistently showed that organizations utilizing analytical and reporting tools are better equipped to identify business trends, monitor operational performance, allocate resources effectively, and improve competitiveness. These

findings highlight the value of integrating reporting and analytics features into business information systems.

A common theme identified across all reviewed studies is the positive impact of automation on organizational performance. Regardless of the specific business function being addressed, the studies consistently reported improvements in efficiency, accuracy, transparency, accountability, and decision-making. The findings suggest that integrating digital technologies into business operations reduces reliance on manual processes and enables organizations to manage information more effectively.

Despite the numerous benefits identified in the reviewed studies, most existing systems focus on a single organizational function such as sales management, attendance monitoring, inventory control, or business analytics. Only a limited number of studies explored the integration of multiple business processes into a unified platform. This indicates a research gap in the development of comprehensive systems capable of managing several operational functions simultaneously.

Therefore, the present study seeks to address this gap through the development of a Web-Based Sales, Attendance, and Inventory Management System for Mr. Softy. Unlike existing systems that focus on individual functions, the proposed system integrates sales processing, attendance monitoring, inventory management, and reporting features into a centralized web-based platform. Through this integration, the system aims to improve operational efficiency, ensure accurate record management, strengthen employee accountability, enhance inventory control, and provide reliable information for managerial decision-making. Ultimately, the proposed system is expected to contribute to the overall productivity, effectiveness, and growth of the organization.

CHAPTER 3

METHODOLOGY

Research Design

This study uses a developmental research design in creating a Web-Based Mr. Softy Integrated Business Intelligence Platform. This research design is suitable because the study focuses on the design, development, and evaluation of a system that will help improve the business operations of Mr. Softy. The system aims to solve problems related to manual inventory checking, manual sales recording, and manual attendance tracking.

The researchers will first gather information about the current process and identify the problems experienced by the business. After identifying the system requirements, the researchers will design and develop the web-based platform using appropriate programming languages, web technologies, and database management tools. The developed system will include features for inventory management, sales monitoring, attendance tracking, and report generation.

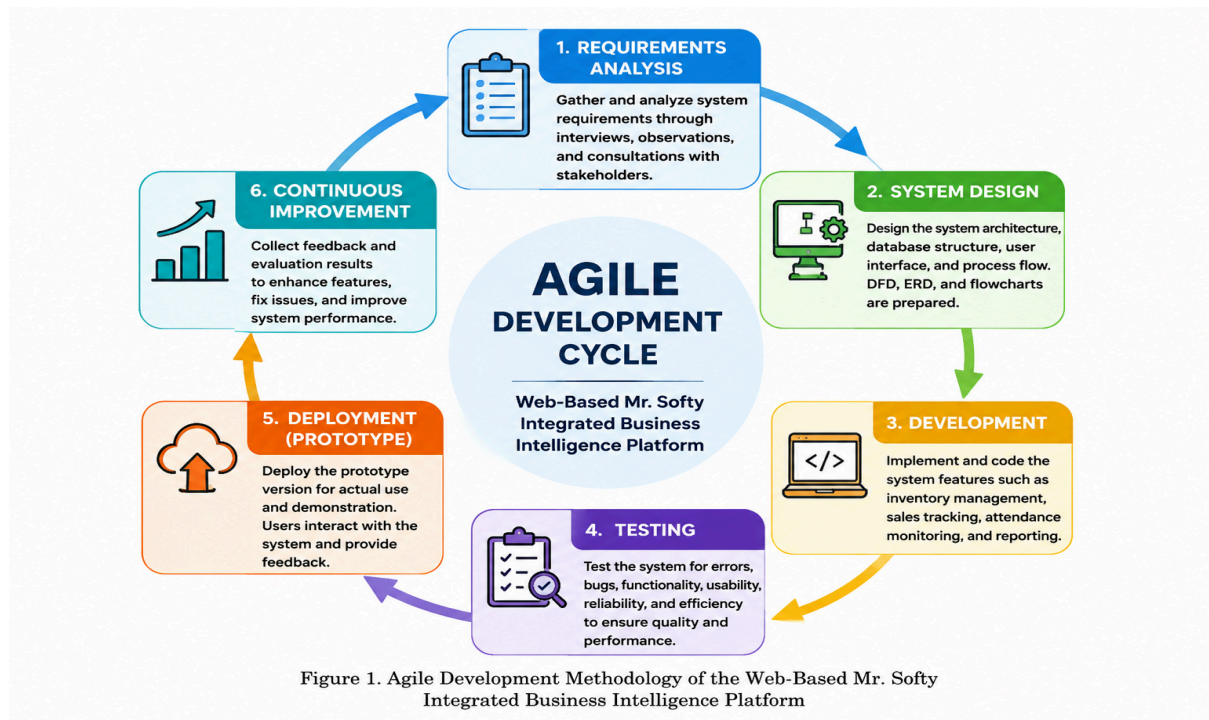
After the development process, the system will be tested and evaluated based on functionality, usability, efficiency, and reliability to determine if it meets the needs of the business. The evaluation results will also be used to improve the system and ensure that it provides a more efficient and accurate way of managing the operations of Mr. Softy.

System Development Methodology

This study will utilize the **Agile Software Development Methodology** in the development of the **Web-Based Mr. Softy Integrated Business Intelligence Platform**. Agile is a widely used software development approach that emphasizes flexibility, collaboration, continuous improvement, and rapid delivery of functional system components. This methodology is suitable for the study because it allows developers to adapt to changing requirements and incorporate user feedback throughout the development process, ensuring that the final system effectively addresses the needs of the business.

Unlike traditional development methodologies that follow a rigid sequence of phases, Agile promotes an iterative and incremental approach. The system will be developed through a series of short development cycles called sprints. Each sprint focuses on the design, development, testing, and evaluation of specific system features. This approach enables the researchers to identify and resolve issues early, improve system functionality continuously, and ensure that project objectives are achieved efficiently.

The development process will consist of the following phases:



Requirements Analysis

In this phase, the researchers will gather and analyze all information necessary for the development of the proposed system. Data gathering techniques such as interviews, observations, and consultations with the business owner, administrator, and staff will be conducted to understand the current business processes and identify existing problems. The researchers will examine how sales transactions, inventory monitoring, and employee attendance are currently managed. The collected data will be analyzed to determine the functional and non-functional requirements of the system, ensuring that the proposed platform addresses the needs of the business and improves operational efficiency.

System Design

After identifying the system requirements, the researchers will proceed with designing the overall structure and functionality of the system. This phase involves creating the user interface design, database architecture, system modules, and process workflows. The researchers will also develop diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), flowcharts, and use case diagrams to visualize how data moves within the system and how users interact with it. The design phase serves as the blueprint for the development process and ensures that all system requirements are properly represented.

Development

During this phase, the researchers will convert the approved system design into a working application through coding and programming. Appropriate programming languages, frameworks, and database management systems will be utilized to develop the platform. Core features such as sales transaction processing, inventory management, employee attendance monitoring, user account management, report generation, and system security will be implemented. The researchers will follow the design specifications to ensure that the developed system meets the identified requirements and business objectives.

Testing

Once the development process is completed, the system will undergo a series of testing procedures to identify and correct errors, bugs, and system defects. Functional testing will be performed to verify that each feature operates according to its intended purpose. Usability testing will assess the ease of use and user-friendliness of the system, while reliability and performance testing will evaluate system stability and efficiency. Any issues discovered during testing will be documented and resolved to ensure the quality and effectiveness of the system before deployment.

Deployment (Prototype)

After successfully passing the testing phase, the prototype version of the system will be deployed for actual use and evaluation. The business owner, administrator, and selected users will be given access to the system to perform real-world tasks such as processing sales, monitoring inventory, and recording employee attendance. This phase allows the researchers to observe how the system performs in a practical environment and gather valuable feedback regarding its functionality, usability, and overall effectiveness.

Continuous Improvement

The final phase focuses on enhancing and refining the system based on the feedback and evaluation results gathered during deployment. Suggestions, concerns, and recommendations from users will be carefully reviewed to identify areas for improvement. Necessary modifications, feature enhancements, bug fixes, and performance optimizations will be implemented to increase system efficiency, accuracy, usability, and reliability. This iterative process ensures that the proposed platform continues to meet the evolving needs of the business and provides long-term value to its users.

SYSTEM ARCHITECTURE

The Mr. Softy Web-Based Integrated Business Intelligence Platform follows a **client-server architecture**. In this type of architecture, the client side allows users to access and interact with the system through a web browser, while the server side is responsible for processing requests, handling business logic, managing the database, and generating reports and analytics.

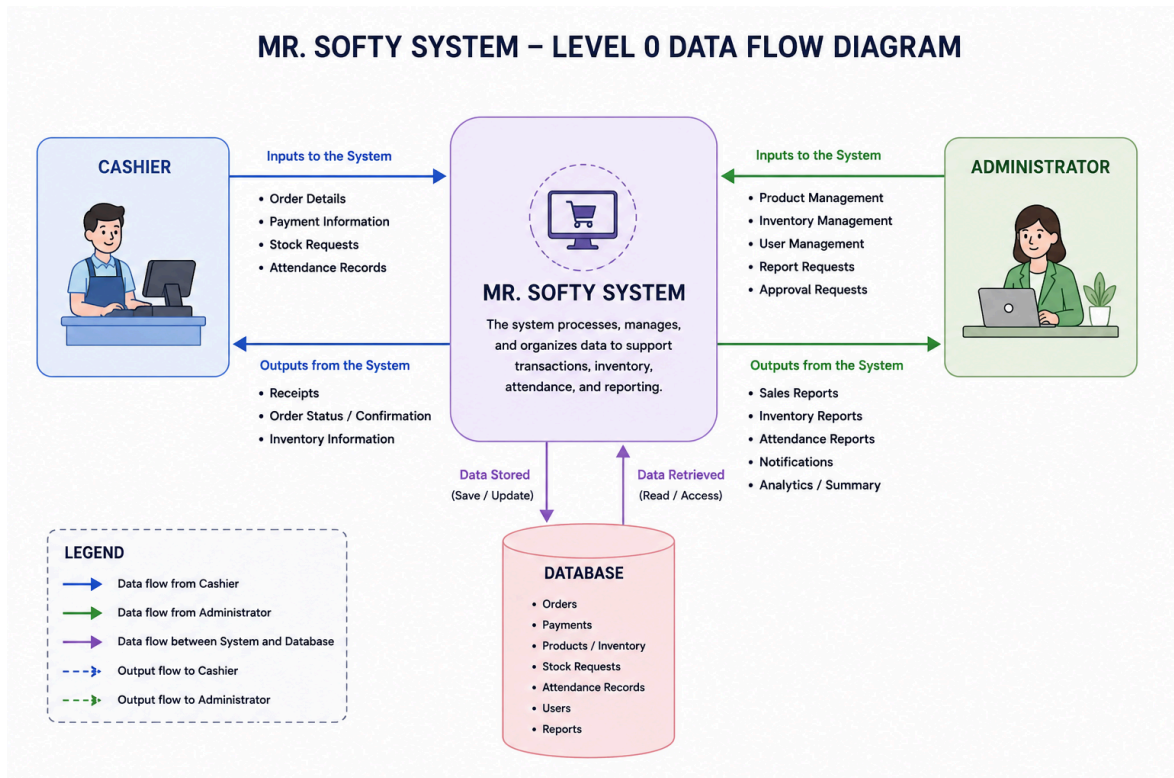
The client sends requests such as recording sales, checking inventory, or monitoring attendance to the server. The server then processes the data and stores it in the database before sending the results back to the user interface. This architecture helps the system provide faster data processing, centralized data management, and secure access for multiple users at the same time.

The **client-server architecture** is suitable for the study because it supports real-time data access, easier system maintenance, and improved scalability. It also allows the researchers to manage and update the system efficiently without affecting the users' access to the platform.

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User Interaction Flow

Data Flow Diagram (LEVEL 0)



The **Level 0 Data Flow Diagram (DFD)** of the **Mr. Softy System** provides a high-level overview of how data flows between the system, its users, and the database. It illustrates the interactions among the **Cashier**, **Administrator**, **Mr. Softy System**, and the **Database**, showing how information is entered, processed, stored, and retrieved to support daily business operations.

The **Cashier** serves as one of the primary external entities of the system. The cashier inputs essential information such as customer order details, payment information, stock requests, and attendance records into the system. These data are processed by the Mr. Softy

System to facilitate transaction processing, inventory monitoring, and attendance tracking. After processing, the system generates outputs including receipts, order confirmations, and inventory information, which are provided to the cashier to support customer service and operational tasks.

The **Administrator** is responsible for managing and overseeing the overall operation of the system. The administrator submits product management updates, inventory management information, user management tasks, report requests, and approval requests to the system. In return, the system provides outputs such as sales reports, inventory reports, attendance reports, notifications, and analytical summaries. These outputs assist the administrator in monitoring business performance, making informed decisions, and ensuring efficient store operations.

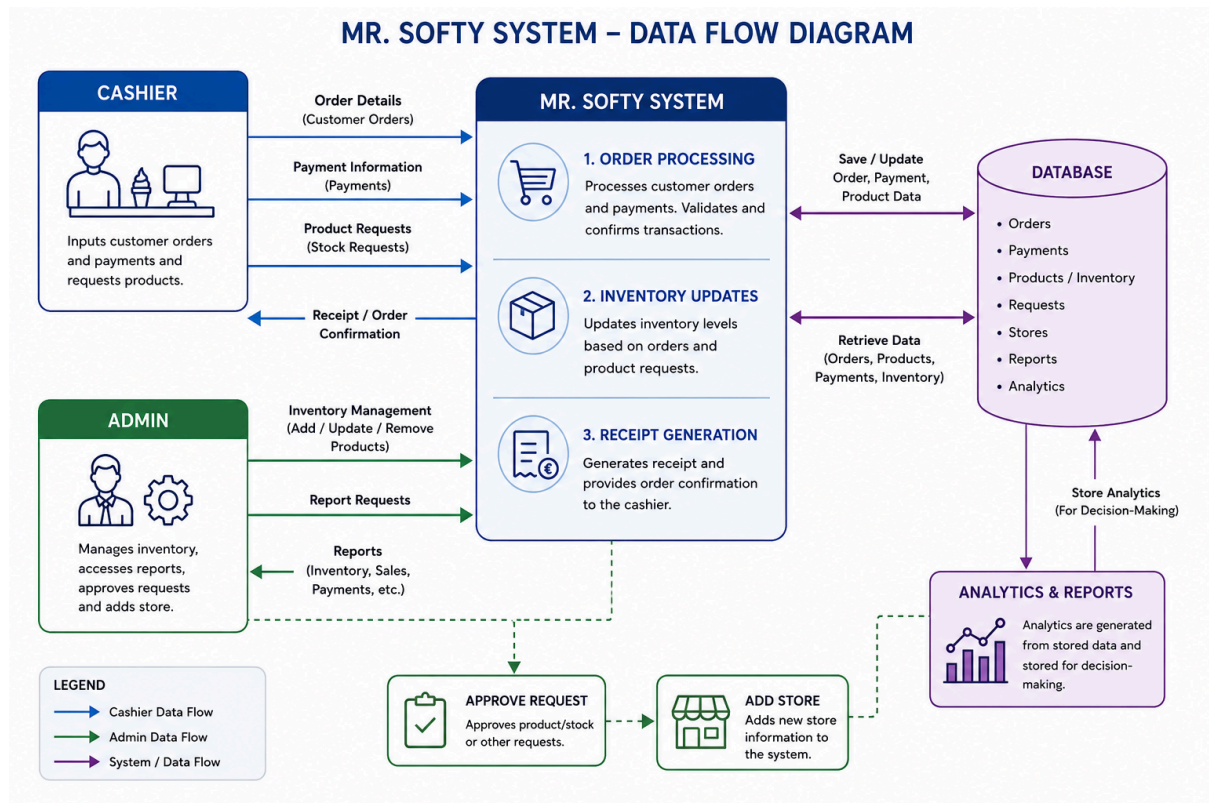
At the center of the diagram is the **Mr. Softy System**, which acts as the main processing unit. It receives data from both the cashier and administrator, processes the information, and performs various functions related to sales transactions, inventory management, attendance monitoring, reporting, and record management. The system ensures that all business activities are organized and accurately recorded.

The **Database** functions as the central repository of information. It stores important records such as orders, payments, products and inventory data, stock requests, attendance records, user information, and reports. The system continuously interacts with the database by saving, updating, and retrieving data whenever necessary. This centralized storage ensures data consistency, security, and accessibility across all system functions.

Overall, the Level 0 Data Flow Diagram demonstrates how the Mr. Softy System integrates sales processing, inventory management, attendance tracking, reporting, and

administrative functions into a unified computerized platform. By enabling seamless data flow between users and the database, the system enhances operational efficiency, improves record accuracy, supports effective decision-making, and reduces the limitations associated with manual business processes.

Data Flow Diagram (LEVEL 1)



The **Level 1 Data Flow Diagram (DFD)** of the Mr. Softy System provides a detailed representation of how data flows between the system's users, processes, and database. It breaks down the major functions of the system into specific processes, showing how information is collected, processed, stored, and utilized to support daily business operations. The primary external entities involved in the system are the Cashier and the Administrator, while the database serves as the central repository of information.

The first process, **Order Processing**, handles customer transactions entered by the cashier. The cashier submits order details, payment information, and product requests through the system. The Order Processing module validates the information, calculates transaction totals, confirms payments, and records customer purchases. Once the transaction is successfully

completed, the system generates an order confirmation and sends the corresponding transaction data to the database for storage.

The second process, **Inventory Updates**, manages inventory-related activities. After each successful transaction, the system automatically updates inventory records by deducting the quantity of products sold from the available stock. Product requests submitted by the cashier are also recorded and processed within this module. The Inventory Updates process retrieves existing inventory data from the database and saves any modifications to ensure that stock information remains accurate and up to date. This process helps prevent stock shortages, inventory discrepancies, and manual errors in inventory monitoring.

The third process, **Receipt Generation**, is responsible for producing transaction receipts and order confirmations. After the order has been processed and payment has been verified, the system generates a receipt containing transaction details such as purchased items, quantities, prices, total amount, and payment information. The receipt is then provided to the cashier for distribution to the customer. This process ensures proper documentation of every transaction and serves as proof of purchase.

The **Database** acts as the central data storage component of the system. It stores and maintains important records, including customer orders, payment transactions, inventory data, product information, stock requests, store information, reports, and analytics. The database continuously exchanges information with the system processes by receiving newly generated records and providing existing data whenever retrieval is required. This centralized storage ensures data consistency, security, and accessibility throughout the system.

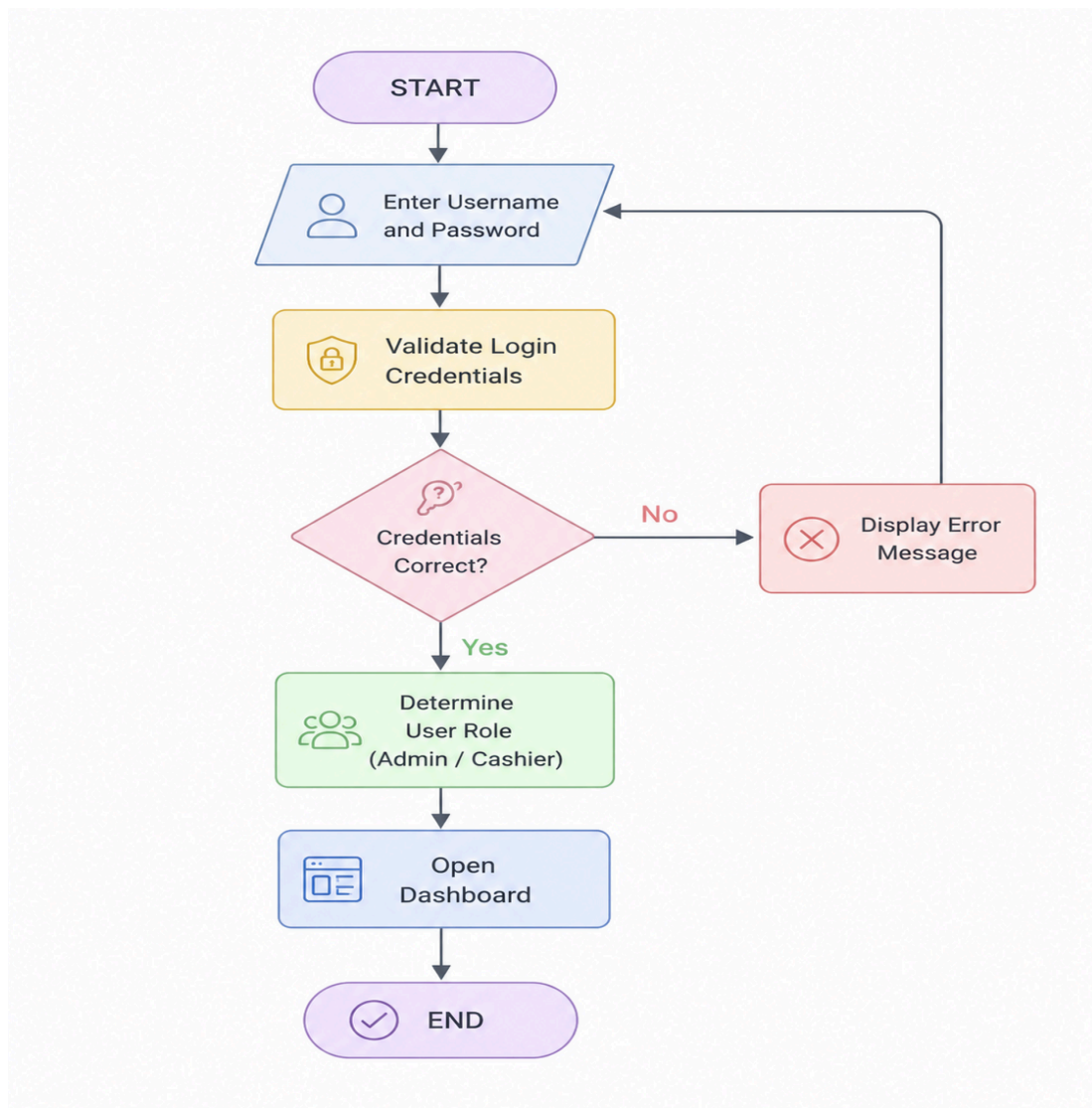
The **Administrator** plays a crucial role in managing system operations. The administrator submits inventory management requests, such as adding, updating, or removing products from the inventory database. The administrator can also request reports, monitor sales and inventory activities, approve stock requests, and manage store information. Through these functions, the administrator ensures that inventory levels are properly maintained and business operations run smoothly.

The **Approve Request** function allows the administrator to review and authorize stock or product requests submitted through the system. Once approved, the corresponding updates are recorded in the database and reflected in inventory records. This process helps maintain proper inventory control and ensures that stock requests follow the organization's approval procedures.

The **Add Store** function enables administrators to register and manage store information within the system. Store details are saved in the database and become available for reporting, monitoring, and operational management purposes. This function supports businesses operating in multiple locations by maintaining organized store records.

The **Analytics and Reports** module generates meaningful information from the data stored in the database. Sales records, payment transactions, inventory levels, and operational data are analyzed to produce reports and performance indicators. These reports provide administrators with valuable insights that support decision-making, business planning, inventory forecasting, and performance evaluation. The generated analytics are stored in the database for future reference and reporting purposes.

Overall, the Level 1 Data Flow Diagram illustrates how the Mr. Softy System integrates transaction processing, inventory management, reporting, request approval, and store management into a unified computerized platform. By automating these processes and maintaining a centralized database, the system improves operational efficiency, data accuracy, inventory control, and managerial decision-making while reducing manual work and processing errors.

Flow Chart

This figure presents the Login Process Flowchart of Web-Based Mr. Softy Integrated Business Intelligence Platform. This flowchart illustrates the sequence of activities performed when a user attempts to access the system. It serves as a security mechanism that ensures only authorized users can access the system's features and data.

The process starts when the user opens the system and enters their username and password in the login interface. These credentials are then submitted to the system for verification. The

system validates the entered information by comparing it with the user records stored in the database.

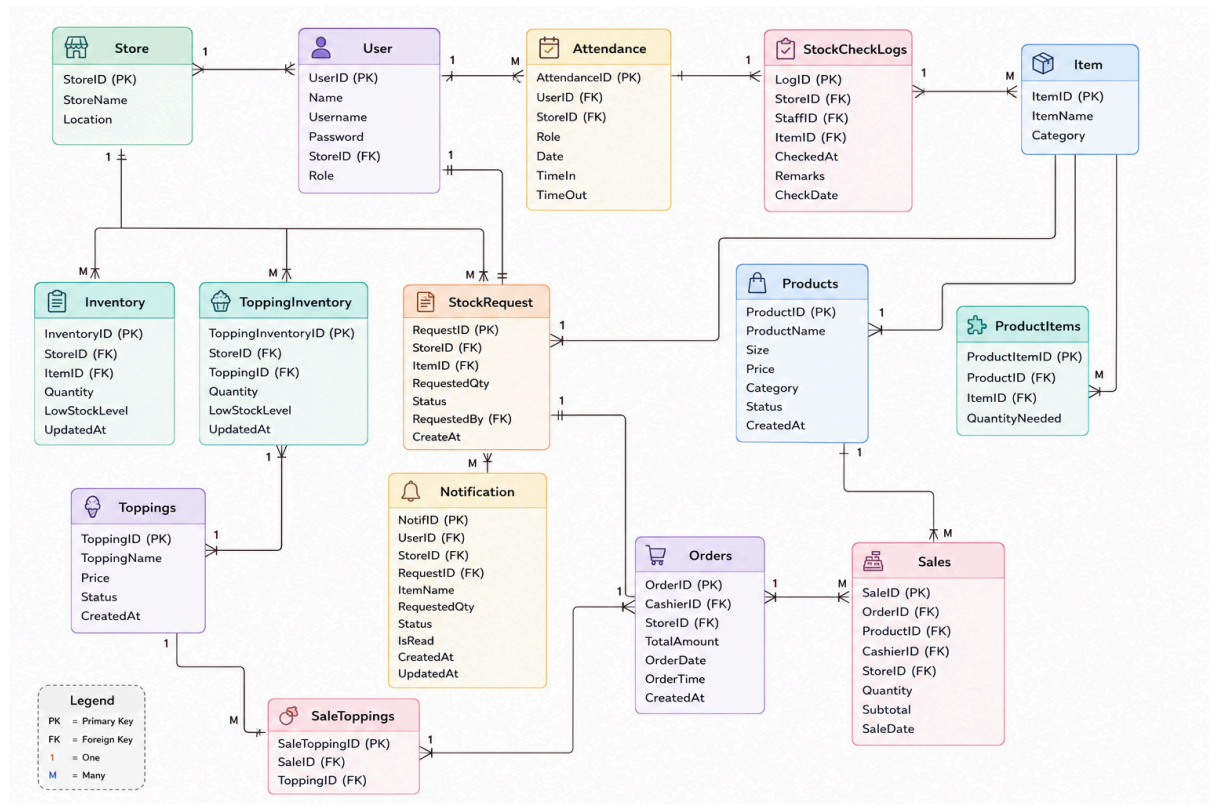
After the validation process, the system evaluates whether the provided credentials are correct. If the username or password does not match any existing record in the database, the system displays an error message informing the user that the login attempt has failed. The user is then redirected back to the login page and prompted to enter the correct credentials before attempting to log in again. This step helps prevent unauthorized access and ensures that only legitimate users can proceed.

On the other hand, if the entered credentials are valid, the system identifies the role assigned to the user. The system supports role-based access control, wherein different users have different permissions and functionalities depending on their designated role. For example, an Administrator is granted access to system management functions such as inventory monitoring, user management, stock requests, attendance records, and report generation, while a Cashier is provided access only to sales transactions and other cashier-related functions.

Once the user's role has been successfully determined, the system grants access and redirects the user to the appropriate dashboard. The dashboard serves as the main interface where users can perform their assigned tasks and interact with the system's modules. After the dashboard is displayed, the login process is considered complete, and the user can begin utilizing the system according to their privileges.

The login process is an essential component of the system because it enhances security, protects sensitive business information, and ensures that users can only access functions relevant to their responsibilities. Through credential validation and role-based authorization, the system maintains data integrity and prevents unauthorized operations within proposed system.

Entity Relationship Diagram



The figure illustrates the database structure of the proposed Point of Sale (POS), Inventory, and Employee Management System. It shows how different entities interact with one another to support daily business operations, including sales processing, inventory monitoring, stock requests, attendance tracking, and product management.

At the center of the system is the **Store** entity, which serves as the primary reference for users, inventory records, sales transactions, stock requests, attendance records, and topping inventories. Each store can have multiple users, inventory items, sales, and operational records.

The **User** entity stores information about system users such as cashiers and administrators. Users are associated with a specific store and can perform various functions including processing sales, requesting stocks, receiving notifications, and recording attendance.

The **Products** entity contains information about menu items offered by the business. Since each product may require multiple inventory items, the **ProductItems** entity serves as a

bridge table that defines the ingredients and quantities needed for each product. This establishes a many-to-many relationship between products and inventory items.

The **Inventory** entity manages the stock levels of raw materials and supplies used in product preparation. It tracks available quantities, low-stock thresholds, and inventory updates. Similarly, the **Toppings** and **ToppingInventory** entities manage available toppings and their corresponding stock quantities for each store.

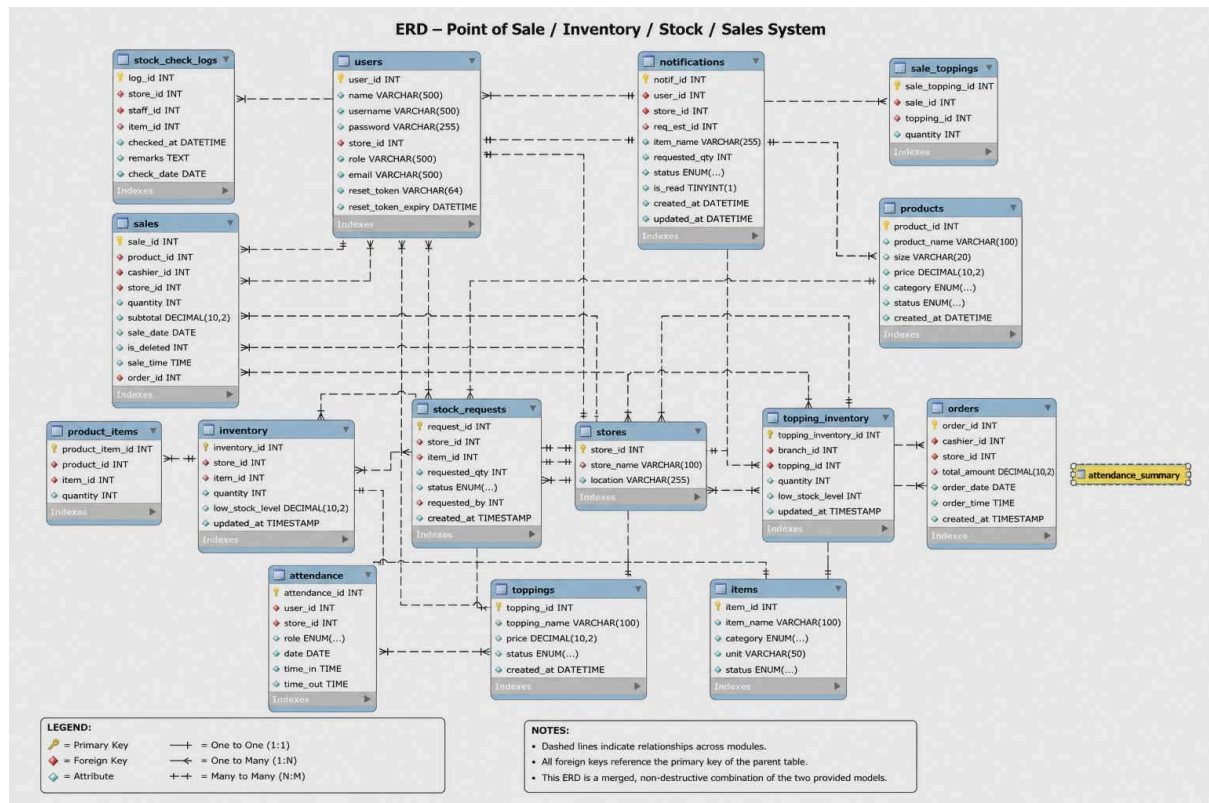
Sales transactions are handled through the **Orders** and **Sales** entities. An order represents a customer's purchase, while sales records contain the individual products included in that order. Additional toppings selected by customers are recorded through the **SaleToppings** entity, which links toppings to specific sales transactions.

To ensure proper inventory management, the **StockRequest** entity records requests for additional stock items made by staff members. These requests generate corresponding entries in the **Notification** entity, allowing administrators to monitor and approve stock replenishment requests.

The **StockCheckLogs** entity maintains records of inventory inspections conducted by staff, including the date, time, and remarks regarding stock conditions. Meanwhile, the **Attendance** entity records employee attendance information such as time-in and time-out logs.

Overall, the ERD demonstrates how the system integrates sales, inventory control, employee management, stock monitoring, and notification processes into a centralized database structure, ensuring efficient and accurate business operations.

Database Design



The database design presents the structure of **Web-Based Mr. Softy Integrated Business Intelligence Platform**. It is composed of several interconnected tables that work together to store, manage, and process business transactions, inventory records, employee attendance, and administrative activities. The design follows a relational database model where each table represents a specific entity and relationships are established through primary and foreign keys to ensure data integrity and consistency.

At the center of the database is the **users** table, which stores information about system users such as administrators and cashiers. This table contains login credentials, user roles, and contact information, allowing the system to implement role-based access control. The users

table is connected to several modules including sales, attendance, stock requests, notifications, and stock check logs.

The **products** table contains information about the products being sold, including product names, sizes, prices, categories, and status. Since products are composed of different ingredients or materials, the **product_items** table serves as a bridge between products and inventory items. This relationship allows the system to determine which inventory items are consumed whenever a product is sold.

The **items** table stores the list of raw materials and inventory supplies used in the business. These items are monitored through the **inventory** table, which records the available quantity of each item per store, the low-stock threshold, and the last update date. This enables the system to automatically monitor inventory levels and identify items that require replenishment.

To support multiple store branches, the **store** table contains branch information such as store name and location. Many operational tables, including inventory, sales, stock requests, attendance, and notifications, reference the store table to identify where a transaction or activity occurred.

The **orders** table records customer orders and serves as the main transaction record during sales operations. Related to this, the **sales** table stores detailed sales information such as product sold, cashier assigned, quantity purchased, subtotal amount, sale date, and sale time. This table provides the basis for generating sales reports and monitoring business performance.

To accommodate customizable products, the system includes a **toppings** table that stores available toppings and their corresponding prices. The **sale_toppings** table acts as a junction table that links sales transactions with selected toppings. Meanwhile, the **topping_inventory** table manages the stock levels of topping ingredients for each branch, ensuring that topping availability is accurately monitored.

Inventory management is further supported by the **stock_requests** table. This table records requests for additional inventory items whenever stock levels become insufficient. Each request contains information about the requested item, quantity, requesting user, status, and creation date. To keep users informed, the **notifications** table stores system-generated notifications related to stock requests and inventory concerns.

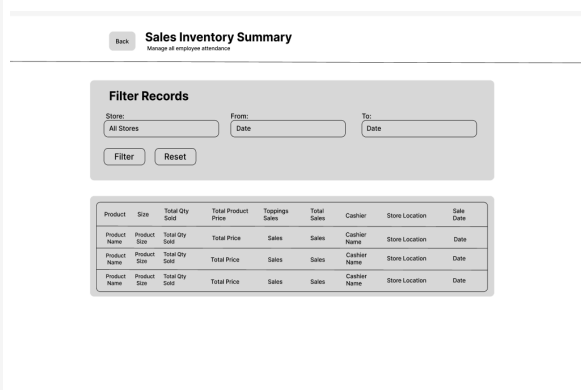
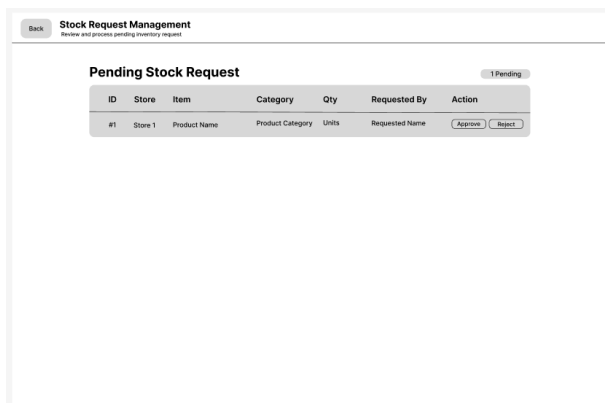
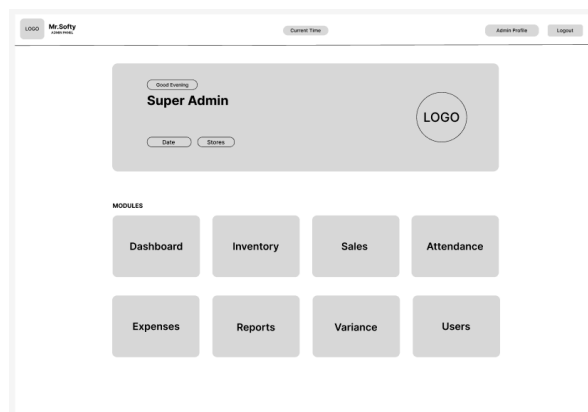
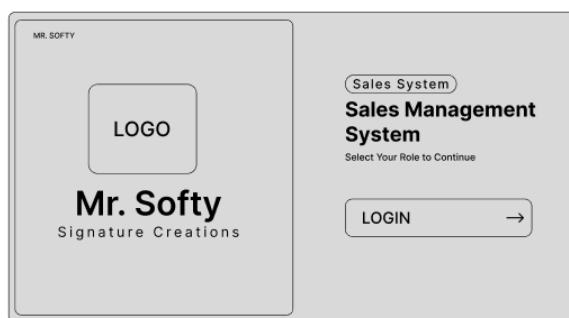
The system also includes the **stock_check_logs** table, which records inventory inspection activities performed by staff members. This table stores information about the employee who conducted the stock check, the item inspected, the store involved, and remarks regarding the inspection. These logs help maintain inventory accuracy and accountability.

For workforce management, the **attendance** table records employee attendance information, including time-in, time-out, date, assigned role, and store location. This allows administrators to monitor employee attendance and working hours efficiently. The **attendance_summary** view provides a summarized representation of attendance records, making it easier to generate attendance reports and analyze employee performance.

Overall, the database design integrates sales processing, inventory monitoring, stock management, attendance tracking, and administrative functions into a single relational database structure. Through the use of well-defined relationships and normalized tables, the

design ensures efficient data storage, minimizes redundancy, maintains data integrity, and supports the operational requirements of the proposed system.

Wire Frame (Admin)



[Back](#)**Sales Inventory Summary**

Manage all employee attendance

Total Records

Today's Date

Filter's Active

Current Time

Filter Records

Start Date:

Date

End Date:

Date

Role:

All Roles

Store ID

Enter Store ID

Filter

Reset

Attendance Records

3 Records

Date	Full Name	Role	Store ID	Location	Time In	Time Out	Action
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete

[Back](#)**Sales Inventory Summary**

Manage all employee attendance

Filter Records

Store:

All Stores

From:

Date

To:

Date

Filter

Reset

Product	Size	Total Qty Sold	Total Product Price	Toppings Sales	Total Sales	Cashier	Store Location	Sale Date
Product Name	Product Size	Total Qty Sold	Total Price	Sales	Sales	Cashier Name	Store Location	Date
Product Name	Product Size	Total Qty Sold	Total Price	Sales	Sales	Cashier Name	Store Location	Date
Product Name	Product Size	Total Qty Sold	Total Price	Sales	Sales	Cashier Name	Store Location	Date

[Back](#)**Attendance Records**

Manage all employee attendance

Total Records

Today's Date

Filter's Active

Current Time

Filter Records

Start Date:

Date

End Date:

Date

Role:

All Roles

Store ID

Enter Store ID

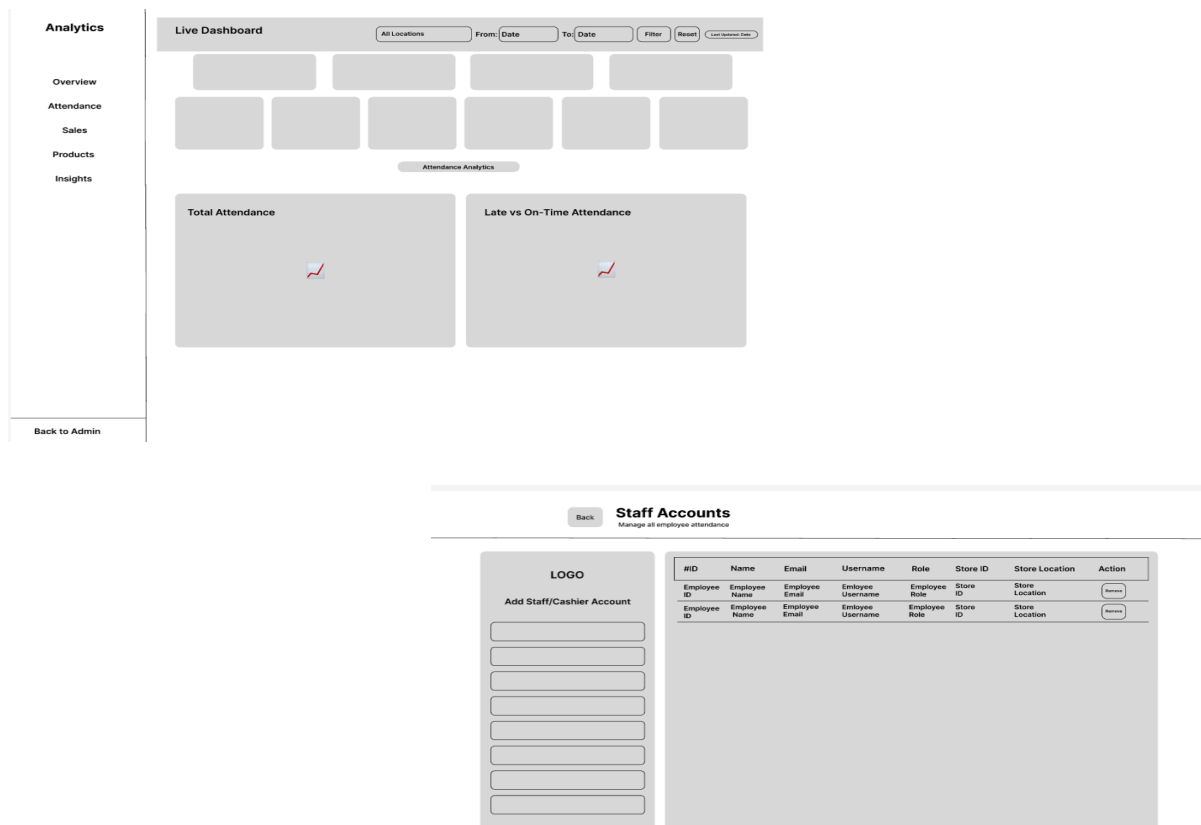
Filter

Reset

Attendance Records

3 Records

Date	Full Name	Role	Store ID	Location	Time In	Time Out	Action
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete
Attendance Date	Employee Name	Employee Role	Store ID	Store Location	Time	Time	Delete



The wireframe presents the proposed design of the Administrator Module for the Web-Based Integrated Business Intelligence platform of Mr. Softy. It serves as a visual representation of the system's interface and shows how users will interact with the different features of the system. The wireframe was created to provide a clear guide during the development process and to ensure that all required functionalities are included.

Login Page

The Login Page is the first interface displayed when users access the system. It contains the company logo, system name, and login fields where administrators and authorized personnel can enter their credentials. This page helps secure the system by preventing unauthorized access to business information and records.

Administrator Dashboard

The Dashboard serves as the main page of the system after a successful login. It provides quick access to the major modules of the system, including Inventory, Sales, Attendance, Expenses, Reports, Variances, and Staff Accounts. The dashboard allows administrators to navigate easily between different functions and manage business operations more efficiently.

Pending Stock Request Module

The Pending Stock Request module displays stock requests submitted by employees. It shows information such as the item requested, quantity, requester, date submitted, and request status. Through this module, administrators can review and approve stock requests to ensure that inventory levels are properly maintained.

Sales Inventory Summary

The Sales Inventory Summary module provides an overview of sales and inventory records. It includes filtering options that allow administrators to search records based on specific dates and categories. The information is presented in a table format for easier monitoring of stock levels, product movement, and sales performance. This feature helps the business track inventory accurately and avoid stock shortages.

Attendance Records Module

The Attendance Records module is used to monitor employee attendance. It displays attendance details such as employee name, role, assigned branch, location, time-in, and time-out. The module also includes filter options to search attendance records based on selected criteria. This feature allows administrators to monitor employee attendance and maintain accurate attendance records.

Analytics Dashboard

The Analytics Dashboard provides visual representations of important business data. It uses charts and graphical reports to display information related to sales performance, attendance records, and inventory status. This module helps administrators analyze trends and make informed decisions based on the collected data.

Staff Accounts Management

The Staff Accounts Management module allows administrators to create, update, and manage user accounts within the system. It contains fields for entering employee information and displays a list of existing accounts. This feature helps ensure that only authorized users can access the system and perform their assigned tasks.

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Wire Frame (Cashier)

Cashier Dashboard — Wireframe

Dashboard

Attendance

New Sale

Sales History

Inventory

Stock Check

Welcome Back, Rhenwel!

Start your day with a smile at Mr.Softy!

Store

Today

TODAY'S TRANSACTIONS

0

Sales recorded today

Live

TODAY'S REVENUE

P0.00

Total collected today

Good

LOW STOCK ITEMS

0

Need restock

Done

ATTENDANCE

Clocked Out

Out at 08:34 PM

Quick Actions

Time In / Out

Record your attendance

New Sale

Process a transaction

Sales History

View past transactions

Manage Inventory

Request stock replenishment

Stock Check

Verify inventory levels

Recent Transactions

View All

No transactions recorded yet today.

Today's Tasks

Morning Stock Check

Verify stock counts and toppings inventory

Done

Process Sales

Handle customer transactions accurately

Pending

End-of-Day Report

Submit daily sales summary before closing

Pending

Freezer Temperature Check

Ensure all storage units are at correct temperature

Pending

Attendance System — Wireframe

Attendance System

Track your work hours

Store #5
Rhenwel

Today's Date

June 8, 2026

08:44:22 PM

All Done!

Attendance completed for today

Time In

08:33 PM

Time Out

08:34 PM

Total Hours

0 hrs 0 mins

Status

Completed

Role

Cashier

Attendance History

Last 10 Records

Date	Name	Time In	Time Out	Total Hours
Jun 8, 2026	John Rhenwel Jerez	08:33 PM	08:34 PM	0 hrs 0 mins

Add Sale — Wireframe

Store #5
Rhenwei

Sales
Start recording sale

Choose a Product [View Order](#)

SIGNATURE CREATIONS

 Choco Glaze Regular ☐ NO CHOCOLATE SYRUP Large ☐ NO CHOCOLATE SYRUP 0 Add to Cart	 Rainbow Delight Regular ☐ NO RAINBOW SPRINKLES Large ☐ NO RAINBOW SPRINKLES 0 Add to Cart	 Cookie Monster Regular ☐ NO COOKIE CRUMBS Large ☐ NO COOKIE CRUMBS 0 Add to Cart	 Nutty Salted Caramel Regular ☐ NO CARAMEL SAUCE Large ☐ NO CARAMEL SAUCE 0 Add to Cart
 Triple Choco Regular ☐ NO TRIPLE CHOCOLATE MIX Large ☐ NO TRIPLE CHOCOLATE MIX 0 Add to Cart	 Sundae Overload Regular ☐ NO SUNDAE TOPPING MIX Large ☐ NO SUNDAE TOPPING MIX 0 Add to Cart	 Kibble Small ☐ NO WAFFLE CONE 0 Add to Cart	 Glant Small ☐ NO WAFFLE CONE 0 Add to Cart

Inventory Management — Wireframe

Store #5

Inventory Management
Monitor stock levels and request replenishments

15 Total Items 0 Sufficient Stock 15 Low Stock Items

INVENTORY ITEMS

ITEM	CATEGORY	CURRENT QTY	STATUS	REQUEST STOCK
BASE				
☐ Ice Cream Base	Base	0	Low Stock	Qty Request
☐ Triple Chocolate Mix	Base	0	Low Stock	Qty Request
CONTAINER				
☐ Paper Cup (Large)	Container	0	Low Stock	Qty Request
☐ Paper Cup (Regular)	Container	0	Low Stock	Qty Request
☐ Waffle Cone	Container	0	Low Stock	Qty Request
FLAVORING				
☐ Vanilla Extract	Flavoring	0	Low Stock	Qty Request
TOPPING				
☐ Caramel Sauce	Topping	0	Low Stock	Qty Request
☐ Chocolate Chips	Topping	0	Low Stock	Qty Request

Morning Stock Check — Wireframe

Store #5

Morning Stock Check
Verify physical stock levels — tick each item once confirmed

15 Total Items 0 Sufficient Stock 15 Needs Restock

NEEDS ATTENTION ☐ **toggle**

☐ Ice Cream Base	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Triple Chocolate Mix	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Paper Cup (Large)	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Paper Cup (Regular)	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Waffle Cone	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Vanilla Extract	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Caramel Sauce	Out of stock Threshold: 5.00	Low Stock ☐ Check
☐ Chocolate Chips	Out of stock Threshold: 5.00	Low Stock ☐ Check

0 / 15 verified

[Request Restock](#) [Submit Check](#)

Update Profile — Wireframe

Cashier Panel
Point of Sale System

Rhenwel
Cashier Role

- Dashboard
- Attendance
- + New Sale
- Sales History
- Inventory
- ✓ Stock Check

→ Logout

← Back to Dashboard

Rhenwel

☐ **PERSONAL INFORMATION**

Full Name
John Rhenwel Jerez

Username
Rhenwel

Email Address
jerezrhenwel@gmail.com

☐ **CHANGE PASSWORD** (leave blank to keep current)

New Password
Min. 6 characters

Confirm New Password
Repeat new password

☐ **CONFIRM IDENTITY**

Current Password *

Required to save changes

Save Changes

The wireframe presents the proposed design of the Cashier Module for the Web-Based Integrated Business Intelligence Platform of Mr. Softy. It serves as a visual representation of the system's interface and demonstrates how cashiers will interact with the different functions of the system. The wireframe was designed to ensure that the system is user-friendly, organized, and capable of supporting daily business operations such as order processing, attendance monitoring, inventory management, and profile maintenance.

Cashier Dashboard

The Cashier Dashboard serves as the main interface after a successful login. It provides an overview of important information, including current sales, order transactions, stock status, and quick access to system features. The dashboard contains navigation menus that allow cashiers to access modules such as Attendance, Add Orders, Inventory Management, Stock

Checking, and Profile Settings. This interface helps users perform their tasks efficiently by providing all necessary tools in one centralized location.

Attendance Monitoring Module

The Attendance Monitoring Module allows cashiers to record and monitor their attendance. The interface displays the current date and time and provides options for time-in and time-out transactions. Attendance records are stored and displayed in a table for monitoring purposes. This feature helps maintain accurate attendance records and supports employee monitoring within the business.

Add Orders Module

The Add Orders Module enables cashiers to process customer orders. It displays available products in a card-based layout where users can select items and add them to the order list. Product information such as name, price, and category is shown to help cashiers identify products quickly. This module simplifies the ordering process and reduces errors during transaction processing.

Inventory Management Module

The Inventory Management Module allows cashiers to view available inventory items and monitor stock levels. The interface displays inventory information in a tabular format,

including product names, categories, quantities, and stock status. Search and filtering options are provided to help users locate specific items easily. This module assists in maintaining accurate inventory records and ensures that products remain available for customer orders.

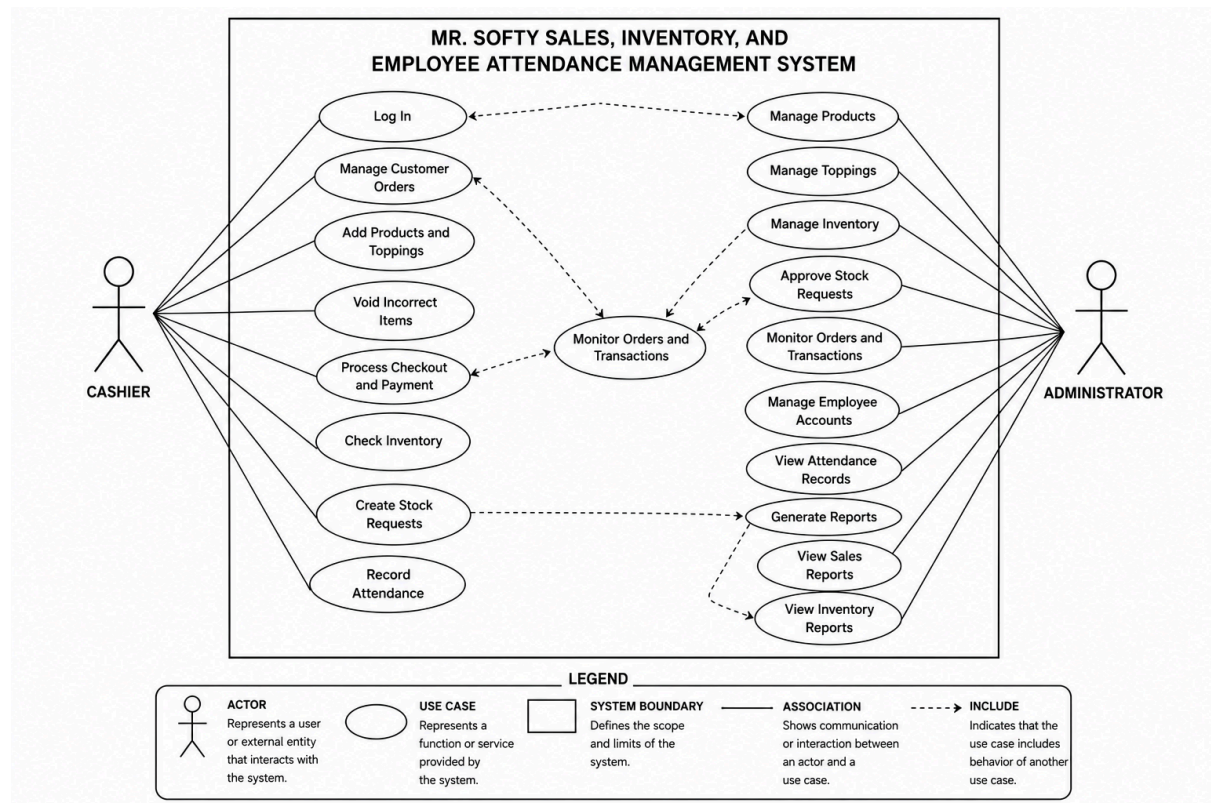
Morning Stock Check Module

The Morning Stock Check Module is used to verify inventory levels at the beginning of each business day. Cashiers can review available stocks and record the current quantity of each item. This process helps identify shortages, monitor stock conditions, and ensure that products are ready for daily operations. The module also supports inventory accuracy by allowing regular stock verification.

Update Profile Module

The Update Profile Module allows cashiers to manage and update their personal account information. Users can modify details such as name, username, password, contact information, and other account settings. This feature helps maintain accurate employee records and allows users to manage their own account information securely.

USE CASE DIAGRAM



The Use Case Diagram illustrates the interactions between the users and the proposed **Mr. Softy Sales, Inventory, and Employee Attendance Management System**. It identifies the different functions of the system and shows how each user interacts with those functions. The diagram helps define the scope of the system and serves as a guide during the system development process.

The system has two primary actors: the **Cashier** and the **Administrator**. These actors are located outside the system boundary because they represent the users who interact with the system.

The **Cashier** is responsible for handling daily business operations. The cashier can log in to the system, manage customer orders, add products and toppings, void incorrect items, process customer payments, check inventory availability, create stock requests, and record

employee attendance. These functions help ensure that sales transactions are processed efficiently and that inventory information remains updated.

On the other hand, the **Administrator** has higher-level management responsibilities. The administrator can manage products, toppings, and inventory records, approve stock requests submitted by the cashier, monitor orders and transactions, manage employee accounts, view attendance records, generate reports, and review sales and inventory reports. These functions allow the administrator to supervise business operations and make informed decisions based on system-generated data.

The diagram also shows the relationships between use cases through the use of «**include**» and «**extend**» relationships. The «**include**» relationship indicates that one use case requires the functionality of another use case as part of its process. For example, generating reports may include viewing sales reports and inventory reports. Meanwhile, the «**extend**» relationship indicates optional or additional functionality that may occur under specific conditions, such as voiding incorrect items while managing customer orders.

The **system boundary** represents the scope of the proposed system and contains all the use cases that the system can perform. The actors interact with these use cases through **association relationships**, which are represented by solid lines connecting the actors to the use cases.

Overall, the Use Case Diagram provides a visual representation of the functional requirements of the proposed system. It helps the researchers, developers, and stakeholders understand how users interact with the system and ensures that all necessary features are identified before the implementation phase begins.

Tools and Technologies Used

The development of the proposed system utilized various software tools and technologies to ensure efficient system design, implementation, and database management. These technologies were selected based on their reliability, accessibility, and compatibility with web-based applications.

Frontend Technologies

HTML (HyperText Markup Language)

HTML serves as the foundation of the system's user interface. It is used to structure and organize the content displayed on web pages, including forms, buttons, tables, navigation menus, and other interactive elements. Through HTML, users can easily interact with the system to perform tasks such as managing orders, viewing inventory, recording attendance, and generating reports.

CSS (Cascading Style Sheets)

CSS is utilized to enhance the visual appearance and layout of the system. It controls the design aspects of the application, including colors, fonts, spacing, alignment, and responsiveness. By using CSS, the system provides a more user-friendly and professional interface that improves usability and user experience for both cashiers and administrators.

JavaScript

JavaScript is used to add interactivity and dynamic functionality to the system. It enables features such as form validation, real-time calculations, automatic updates, confirmation messages, and responsive user interactions without requiring the page to reload. This improves the efficiency and responsiveness of the application.

Backend Technologies

PHP (Hypertext Preprocessor)

PHP serves as the server-side scripting language responsible for processing user requests and managing system operations. It handles data transactions between the user interface and the database, including user authentication, order processing, inventory updates, attendance recording, and report generation. PHP enables the system to execute business logic securely and efficiently.

MySQL

MySQL is the relational database management system used to store, organize, and manage the system's data. It maintains records related to products, toppings, inventory, sales transactions, stock requests, employee attendance, and user accounts. MySQL ensures data consistency, security, and efficient retrieval of information required by the system.

Database Management Tool

MySQL Workbench

MySQL Workbench is utilized as the database design and administration tool for the project. It assists developers in creating and managing the database structure through Entity-Relationship Diagrams (ERDs), designing tables, defining relationships, executing SQL queries, and monitoring database performance. The tool helps ensure that the database is properly organized and optimized for the system's requirements.

Development Environment

Visual Studio Code (VS Code)

Visual Studio Code is the primary Integrated Development Environment (IDE) used during system development. It provides a lightweight yet powerful coding environment with features such as syntax highlighting, code completion, debugging tools, and extension support for PHP, HTML, CSS, JavaScript, and MySQL. These features improve development efficiency, code quality, and project management.

Summary

The combination of **HTML, CSS, JavaScript, PHP, MySQL, MySQL Workbench, and Visual Studio Code** provides a complete development environment for building the proposed system. These technologies work together to deliver a secure, efficient, and user-friendly web-based application capable of supporting the daily operations of Mr. Softy.

Data Gathering Techniques

To gather the necessary information for the development of the proposed Point-of-Sale, Inventory Management, and Attendance Monitoring System, the researchers employed several data gathering techniques. These techniques were used to understand the current operational processes, identify existing problems, and determine the system requirements needed to address the business's needs effectively.

Interviews

The researchers conducted interviews with the business owner, administrator, and staff members who are directly involved in the daily operations of the business. The interviews aimed to collect detailed information regarding the current procedures for sales transactions, inventory management, employee attendance monitoring, and report generation.

Through face-to-face discussions, the researchers were able to identify the challenges and limitations of the existing manual system, such as errors in record-keeping, delays in report preparation, difficulties in inventory tracking, and inefficiencies in attendance monitoring. The interview process also provided an opportunity for the respondents to share their suggestions and expectations for the proposed computerized system.

Furthermore, the information gathered through interviews served as a primary source in determining the functional and non-functional requirements of the system. By obtaining insights directly from the users, the researchers ensured that the proposed system would address actual operational problems and support the business's daily activities more effectively.

Observation

Observation was utilized to examine and understand the actual workflow and operational procedures of the business in its natural setting. The researchers observed how sales transactions were processed, how inventory records were maintained, how employee attendance was monitored, and how daily reports were prepared and managed.

This technique allowed the researchers to witness firsthand the sequence of activities performed by employees and administrators during business operations. Through observation, the researchers identified inefficiencies, repetitive tasks, and areas where human errors commonly occurred. It also enabled the researchers to verify and validate the information obtained during the interviews.

Additionally, observation provided valuable insights into user behavior, system requirements, and operational challenges that may not have been fully expressed during interviews. By closely examining the actual business processes, the researchers gained a comprehensive understanding of the organization's workflow, which contributed significantly to the design and development of a system that aligns with the business's operational needs and objectives.

Document Analysis

The researchers also reviewed existing business documents and records, including sales receipts, inventory logs, attendance records, stock request forms, and manual reports. These documents served as supporting sources of information that helped the researchers understand the structure, format, and content of the data currently being used by the business.

Through document analysis, the researchers identified the types of information that needed to be stored, processed, and generated by the proposed system. This technique also helped ensure that the computerized system would maintain consistency with existing business records while improving accuracy, accessibility, and efficiency in data management.

Testing Procedures

To ensure the quality, reliability, and effectiveness of the proposed Inventory, Sales, and Attendance Monitoring System, the researchers will conduct several levels of testing throughout the development process. These testing procedures will help identify and resolve errors, verify system functionality, and ensure that the system meets the requirements of the business.

- **Unit Testing**

Unit Testing will be performed to evaluate individual modules or components of the system separately. Each feature, such as login authentication, inventory management, sales transaction processing, attendance recording, and report generation, will be tested independently to ensure that it functions correctly according to the specified requirements. This testing helps identify coding errors and defects at an early stage of development, making them easier and less costly to fix.

- **Integration Testing**

After the individual modules have been tested, Integration Testing will be conducted to verify that the different components of the system work together properly. The researchers will examine the interaction between modules, such as the connection between sales transactions and inventory updates, attendance records and report generation, and user accounts and system access controls. This testing ensures that data flows accurately between modules and that there are no communication issues within the system.

- **System Testing**

System Testing will be performed on the fully integrated system to evaluate its overall functionality, performance, security, and reliability. The researchers will test the complete platform in an environment that closely resembles actual business operations. This process will verify that all system features function as intended and meet the established requirements. The testing will also assess the system's ability to handle multiple transactions, maintain data accuracy, and provide stable performance during regular use.

- **User Acceptance Testing (UAT)**

User Acceptance Testing will be conducted by allowing the business owner, administrator, and selected users to evaluate the system based on actual business tasks and operations. Users will interact with the system and provide feedback regarding its usability, efficiency, functionality, and overall performance. The purpose of this testing is to determine whether the system meets the expectations and needs of the users. Any suggestions, concerns, or issues identified during this phase will be reviewed and addressed before the final implementation of the system.

Evaluation Criteria

The researchers will evaluate the developed system using software quality criteria to determine its effectiveness, performance, and acceptance by users. The evaluation will focus on functionality, usability, reliability, security, and efficiency to ensure that the system meets the operational requirements of the business and provides a satisfactory user experience.

The proposed Inventory, Sales, and Attendance Monitoring System will be evaluated based on several quality criteria to determine its effectiveness, performance, and suitability for business operations. These criteria will serve as the basis for assessing whether the system successfully meets the needs and expectations of its users.

- **Functionality**

Functionality refers to the system's ability to perform its intended tasks accurately and efficiently. The researchers will evaluate whether all features and modules, including sales transaction processing, inventory management, attendance monitoring, report generation, and user account management, function according to the specified requirements. The system must produce correct outputs, process data accurately, and support the daily operations of the business without errors.

- **Usability**

Usability measures how easy and convenient the system is for users to learn, operate, and navigate. The researchers will assess whether the user interface is clear, organized, and user-friendly for cashiers, administrators, and other authorized users. Factors such as ease of navigation, accessibility of features, clarity of instructions, and overall user experience will be considered. A highly usable system allows users to complete tasks efficiently with minimal training and effort.

- **Reliability**

Reliability evaluates the system's ability to perform consistently and accurately over time. The researchers will examine whether the system can operate continuously without frequent failures, crashes, or unexpected errors. The system should maintain data integrity, process transactions correctly, and provide dependable performance during normal business operations. A reliable system ensures that users can trust the platform to perform its functions whenever needed.

- **Security**

Security focuses on protecting the system and its data from unauthorized access, misuse, and potential threats. The researchers will evaluate the effectiveness of user authentication, access controls, data validation, and error-handling mechanisms implemented within the system. The system should ensure that sensitive information, such as employee records, sales data, and inventory information, is properly secured and accessible only to authorized users. Input validation will also be assessed to prevent invalid or harmful data from entering the system.

- **Efficiency**

Efficiency refers to the system's ability to perform tasks quickly while utilizing resources effectively. The researchers will assess the speed of transaction processing, report generation, data retrieval, and inventory updates. The system should respond promptly to user actions and maintain smooth performance even when handling multiple operations. Efficient resource utilization helps improve productivity and supports the daily workflow of the business.

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